

Expected vs. Unexpected Extremes: The Origins of the MAX Anomaly

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Abstract

We find that the MAX anomaly is driven by unexpected extremes consistent with temporary mispricing rather than by predictable exposure to tail risk. This paper examines whether the anomaly reflects compensation for expected tail risk or instead arises from unexpected extreme shocks. Using daily returns for NYSE, Amex, and Nasdaq stocks from 1962 to 2023, we forecast next-month maximum daily returns using AR(1), ARMA(1,1), quantile–tail, and EVT-based models. Out-of-sample comparisons show that the ARMA(1,1) specification provides the most accurate forecasts. We then decompose realized MAX into expected and unexpected components and evaluate their ability to explain the cross-section of stock returns. Portfolio sorts and Fama–MacBeth regressions show that expected MAX has little predictive power for future returns once unexpected extremes are considered. In contrast, stocks experiencing unexpectedly large extreme returns earn significantly lower subsequent returns. Double-sort analyses further show that the strongest underperformance occurs when extreme realizations substantially exceed prior expectations, highlighting the central role of unexpected shocks in driving the MAX anomaly.

Keywords: *MAX anomaly; extreme returns; expected MAX; unexpected MAX; tail risk; asset pricing*

1.Introduction

Extreme return realizations play an important role in shaping the cross-section of stock returns. A growing literature documents that stocks experiencing unusually large positive daily returns subsequently earn lower future returns, a phenomenon commonly referred to as the MAX anomaly (Bali, Cakici, and Whitelaw, 2011). This finding suggests that stocks exhibiting large short-term upside realizations subsequently underperform relative to otherwise similar firms. Despite extensive empirical evidence documenting the negative relationship between maximum daily returns and future stock performance, the economic mechanism underlying this anomaly remains an open question.

One prominent interpretation of the MAX anomaly links it to investor preferences for lottery-like payoffs. In this view, stocks that occasionally generate extremely large positive returns resemble lottery tickets and therefore attract speculative demand from investors seeking small probabilities of large gains (Barberis and Huang, 2008; Bali et al., 2011). Such demand can temporarily inflate prices, leading to lower future returns when prices subsequently revert. Related studies also suggest that attention-grabbing extreme returns can draw retail investor interest, reinforcing speculative trading and further amplifying temporary mispricing (Barber and Odean, 2008).

Extreme daily returns may arise from a variety of economic forces, including firm-specific information releases, earnings announcements, liquidity shocks, or broader episodes of heightened volatility. Because the MAX measure captures only the largest daily return within a given month, it aggregates these different forces into a single statistic. Consequently, the measure may reflect both persistent variation in the intensity of extreme price movements and short-lived events that temporarily inflate realized returns. This aggregation makes it difficult to interpret whether the MAX anomaly reflects compensation for risk or temporary mispricing.

A central issue in interpreting the MAX anomaly is that realized extreme returns combine two conceptually distinct components which are a predictable component reflecting the expected intensity of extreme price movements and an unexpected component capturing unanticipated shocks. The existing literature largely relies on realized measures of extreme returns without distinguishing between these two elements. As a result, it remains unclear whether the MAX anomaly reflects compensation for predictable exposure to tail risk or whether it arises from unexpected shocks that generate temporary price pressure and subsequent reversals.

In this paper, we address this issue by decomposing realized maximum daily returns into expected and unexpected components. Our approach builds on the growing literature on volatility forecasting, which emphasizes the importance of separating predictable variation in risk from realized innovations. Using insights from this literature, we develop a comprehensive set of models

designed to forecast the maximum daily return within the next month. These forecasts allow us to isolate the portion of extreme return behavior that can be anticipated based on historical information from the component that represents a genuine surprise.

A related challenge in the literature is that most empirical studies rely exclusively on realized extreme returns when measuring lottery-like characteristics. Realized measures may contain substantial noise and may be strongly influenced by transient shocks. The volatility forecasting literature has long emphasized the importance of distinguishing between predictable variation in risk and realized innovations (Andersen, Bollerslev, Diebold, and Labys, 2003; Barndorff-Nielsen and Shephard, 2002; Bollerslev, Patton, and Quaedvlieg, 2016). Applying similar insights to the study of extreme return realizations may therefore provide a clearer understanding of the forces driving the MAX anomaly.

In this paper, we revisit the MAX anomaly by examining the predictability of extreme stock return realizations. Using a set of forecasting models designed to predict the maximum daily return within the next month, we evaluate the extent to which extreme return behavior can be anticipated using historical information. By isolating the predictable and unpredictable components of extreme returns, our framework allows us to examine whether the MAX anomaly is primarily associated with anticipated tail-risk dynamics or with unexpected extreme realizations that occur after prices have already incorporated available information.

Our empirical analysis yields several main findings. First, extreme return realizations exhibit meaningful predictable dynamics. Among the forecasting approaches we consider, the ARMA(1,1) model consistently provides the most accurate out-of-sample forecasts of maximum daily returns, outperforming alternative models across a range of evaluation metrics. This result indicates that a significant portion of extreme return behavior can be predicted using historical information.

Second, separating realized extreme returns into predictable and unpredictable components provides new insights into the MAX anomaly. Portfolio sorts and cross-sectional regressions show that the negative return premium associated with the traditional MAX measure is not driven by the predictable component of extreme returns. Instead, the anomaly is largely associated with unexpected extreme realizations. Stocks experiencing unexpectedly large maximum daily returns subsequently earn significantly lower returns, whereas the predictable component of extreme returns is not significantly related to future returns.

Third, the results suggest that the MAX anomaly is primarily driven by episodic shocks that temporarily elevate realized extreme returns. Such shocks may reflect firm-specific news events, liquidity shocks, or speculative trading episodes that generate unusually large daily price movements. Because these events are not anticipated *ex ante*, they may attract investor attention and speculative demand, leading to temporary price pressure that subsequently reverses. By separating predictable and unexpected components of extreme return realizations, our framework provides a clearer interpretation of the economic forces underlying the MAX anomaly.

Our paper contributes to several strands of the asset pricing literature. First, it contributes to the literature on lottery-like stocks and the MAX anomaly by providing a framework that distinguishes between predictable and unexpected components of extreme return realizations. While prior studies document that stocks with high realized MAX earn lower future returns, we show that this pattern is primarily driven by unexpected extreme shocks rather than predictable exposure to extreme return risk. Second, our analysis contributes to the literature on forecasting financial risk by extending tools from the volatility forecasting literature to the prediction of extreme return realizations. Finally, our findings contribute to the broader debate on whether anomalies associated with extreme returns reflect compensation for risk or temporary mispricing.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data and the forecasting models used to estimate expected maximum daily returns. Section 4 evaluates the forecasting performance of the models and presents the main empirical results on the cross-sectional return implications of expected and unexpected extreme returns. Section 5 concludes.

2. Related literature

A large body of research documents that investors exhibit a strong attraction to lottery-like securities, which can lead to the overvaluation of such assets and influence their subsequent performance. Barberis and Huang (2008) provide a theoretical explanation for this behavior by applying the framework of cumulative prospect theory developed by Tversky and Kahneman (1992). Under cumulative prospect theory, individuals tend to place excessive weight on events with very small probabilities, particularly those associated with extremely favorable outcomes. Because lottery-like stocks offer a small chance of very large positive returns, this probability-weighting behavior leads investors to favor assets with positively skewed payoff distributions. As a result, securities with strong positive skewness may become overpriced and subsequently generate lower excess returns.

Empirical evidence supporting this idea is provided by Bali et al. (2011), who document a strong negative relationship between the maximum daily return in the previous month (MAX) and stock returns in the following month. The authors interpret MAX as a proxy for lottery demand, arguing that investors are willing to pay a premium for stocks that occasionally generate extremely high returns. When this demand pushes the price of high-MAX stocks above their fundamental value, these stocks tend to experience lower future returns that cannot be explained by standard risk factors. Their findings suggest that MAX captures unique information about investors' demand for lottery-like payoffs and possesses significant explanatory power in the cross-section of expected stock returns.

Subsequent studies extend the analysis of the MAX effect beyond the U.S. market. Walkshäusl (2014) reports similar evidence across eleven countries in the European Monetary Union, finding that stocks with higher MAX values tend to underperform those with lower MAX values. Cheon

and Lee (2014) examine a broad international sample covering forty-four countries grouped by geographic regions and also document that high-MAX stocks generally earn lower subsequent returns. The consistency of these findings across multiple markets indicates that the MAX effect is not limited to the U.S. but represents a widespread phenomenon in global equity markets.

A number of studies further explore the mechanisms underlying the MAX effect. Fong and Toh (2014) show that investor sentiment plays an important role, finding that the negative relation between MAX and future returns becomes stronger during periods of high investor optimism. Their results indicate that the effect is mainly driven by the poor subsequent performance of high-MAX stocks rather than unusually high returns for low-MAX stocks. Zhong and Gray (2016) investigate whether the observed relation reflects compensation for risk or instead results from mispricing. Their evidence supports the mispricing explanation, demonstrating that the MAX effect is most pronounced among stocks that appear to be substantially overpriced. High levels of arbitrage risk limit the ability of investors to take short positions in these securities, allowing mispricing to persist. Conversely, among underpriced stocks, the relationship reverses because arbitrage risk discourages investors from taking long positions in high-MAX stocks.

More recently, Liu et al. (2020) provide evidence that investor demand for lottery-like stocks intensifies around corporate earnings announcements. They measure lottery-like characteristics using several proxies, including MAX, expected idiosyncratic skewness, stock price, jackpot indicators, idiosyncratic volatility, and a composite Z-score. Their results show that lottery-like stocks tend to outperform non-lottery stocks during the five-day period preceding earnings announcements, reflecting increased speculative demand. However, this short-term outperformance is followed by the typical underperformance associated with lottery-like securities.

While the literature on the MAX anomaly has primarily focused on the cross-sectional return implications of realized extreme returns, much less attention has been given to the predictability of extreme return realizations themselves. Understanding whether extreme returns are predictable is important because it affects how the MAX anomaly should be interpreted. If extreme realizations are largely predictable, the anomaly may reflect compensation for persistent tail risk. If, instead, extreme returns primarily arise from unexpected shocks, the anomaly may reflect temporary price pressure associated with investor attention or speculative trading. Our approach builds on the broader volatility forecasting literature, which emphasizes the importance of distinguishing between predictable variation in risk and realized innovations (Andersen, Bollerslev, Diebold, and Labys, 2003; Bollerslev, Patton, and Quaedvlieg, 2016).

The simplest benchmark used in forecasting realized financial risk measures is the martingale model, which assumes that the best forecast of next period's realized measure is its current realization. This benchmark is widely used in studies of volatility and idiosyncratic risk, including Ang et al. (2006) and related work on the idiosyncratic volatility puzzle. Although the martingale model provides a useful baseline, it implicitly assumes full persistence and therefore cannot

capture the mean reversion often observed following unusually large realizations of volatility or extreme returns.

To account for persistence in extreme return behavior while allowing partial mean reversion, we consider autoregressive models. Autoregressive specifications are commonly used in the volatility forecasting literature because financial risk measures tend to exhibit substantial temporal persistence. In particular, first-order autoregressive models provide a parsimonious way to capture the idea that recent realizations of risk contain information about future realizations. However, realized measures constructed from high-frequency data often contain measurement noise, which can lead to more complex dynamics than those captured by a simple AR process.

For this reason, we also consider an ARMA(1,1) specification, which has been shown to perform well in forecasting realized variance and related risk measures. Theoretical and empirical studies in the realized volatility literature demonstrate that when latent volatility follows a persistent process but realized measures are observed with noise, the resulting series often follows an ARMA-type process (Barndorff-Nielsen and Shephard, 2002; Andersen et al., 2003). Consistent with this insight, recent work on expected idiosyncratic volatility finds that ARMA models outperform simpler persistence-based benchmarks when forecasting realized risk measures. Applying similar reasoning to extreme return realizations suggests that ARMA-type dynamics may provide an effective framework for forecasting future MAX values.

In addition to time-series models, we also consider approaches motivated by the statistical properties of extreme observations. One such approach is the quantile–tail approximation, which draws on results from order statistics to approximate expected extreme realizations using high quantiles and dispersion measures of the underlying return distribution. This framework allows the expected maximum return to be approximated using information about the upper tail of the return distribution without imposing a specific dynamic structure on realized maxima.

Finally, we consider a model grounded in extreme value theory (EVT). EVT provides a statistical framework for modeling the distribution of rare, extreme observations and has been widely used in finance to study tail risk and market crashes (Embrechts, Klüppelberg, and Mikosch, 1997; McNeil and Frey, 2000). In particular, the peaks-over-threshold approach models exceedances over a high threshold using the Generalized Pareto Distribution (GPD). This approach focuses directly on the behavior of the tail of the return distribution and is therefore well suited to modeling extreme return realizations.

Together, these models represent distinct approaches to forecasting extreme returns: persistence-based models such as martingale and AR processes, dynamic models that allow temporary shocks to revert such as ARMA specifications, distributional approximations based on tail quantiles, and structural models derived from extreme value theory. By comparing the out-of-sample forecasting performance of these models, our paper provides new evidence on the predictability of extreme stock return realizations and the implications for the MAX anomaly.

3. Data and Forecasting Models

3.1 Data Description

The data for this study are drawn from multiple standard sources widely used in the asset-pricing literature. Our primary dataset consists of common stocks (CRSP share codes 10 and 11) listed on the New York Stock Exchange (NYSE), the American Stock Exchange (Amex), and Nasdaq, covering both financial and nonfinancial firms. Stock-level data are obtained from the Center for Research in Security Prices (CRSP) and span the period from July 1962 to December 2023. The sample begins in July 1962 to align with the conventional start date used in much of the cross-sectional expected-return literature.

To mitigate the influence of penny stocks and microstructure noise, we exclude firm-months in which the stock price is below \$1. CRSP daily return files are used to compute monthly maximum daily returns (MAX) for each firm and to construct several firm-level characteristics, including market beta, volatility measures, and idiosyncratic skewness. CRSP monthly returns are employed to calculate intermediate-term momentum and short-term reversal measures. Trading volume data are used to construct illiquidity proxies, while share prices and shares outstanding are used to compute market capitalization (Size).

Accounting data are obtained from Compustat. Book equity values are used to compute firms' book-to-market ratios. Factor returns for risk adjustment are drawn from established sources: the Fama–French factors are obtained from Kenneth French's data library, while the q-factor model data are obtained from Lu Zhang's website. Investor sentiment measures are taken from Jeffrey Wurgler's website. All variables used in the empirical analysis are defined in detail in Appendix A.

To evaluate risk-adjusted returns, we consider two benchmark asset-pricing models. First, we employ the Fama and French (2015) five-factor model augmented with a momentum factor (FF6). Second, we use the Hou, Xue, and Zhang (2015) q-factor model. The full return sample spans July 1962 to December 2023. Data availability for the FF6 model begins in July 1963, while q-factor data are available from January 1967 onward.

3.2 Forecasting Models for Extreme Daily Returns (Expected MAX)

Accurately forecasting extreme returns is central to understanding the pricing of tail-risk and the cross-sectional return patterns associated with lottery-like stocks. The empirical literature documents that measures of extreme upside movements, most prominently the monthly maximum daily return (MAX), predict cross-sectional stock returns (Bali et al. 2011). Yet, little is known about the extent to which these extreme realizations are *predictable* or whether the documented MAX effect simply reflects investors' reactions to *unexpected* tail shocks. To address this, we construct a comprehensive set of forward-looking forecasts for next-month maximum returns

using complementary approaches drawn from time-series econometrics, tail-risk modeling, and extreme-value theory.

Our modeling philosophy mirrors the variance-forecasting framework in Andersen et al. (2003), Bollerslev et al. (2016), Ghysels et al. (2019), and Nelson (1991). Just as realized variances approximate integrated variances under mild conditions, the monthly MAX statistic belongs theoretically to the class of “block maxima” estimators widely studied in the extreme-value literature (Leadbetter et al., 2012; Embrechts et al., 1997). Because MAX captures the single most extreme daily outcome within a month, its dynamics are inherently shaped by (i) volatility persistence, (ii) jump intensity, and (iii) tail heaviness. Accordingly, we develop four forecasting models grounded in distinct theoretical frameworks: autoregressive (AR), ARMA with measurement noise, quantile-tail approximations for extreme order statistics, and Peaks-over-Threshold (POT) modeling using the Generalized Pareto Distribution (GPD).

Section 3.1 describes the construction of monthly MAX and discusses sampling considerations analogous to microstructure issues in realized-variance research. Section 3.2 outlines the four baseline predictive models. Section 3.3 develops additional motivations and robustness considerations, including the economic intuition underlying each model class.

3.3 Constructing Monthly Maximum Daily Returns

Let $r_{i,d}$ denote the daily return of firm i on day d . For a given month t , the block maximum is defined as:

$$\text{MAX}_{i,t} = \max_{d \in t} \{r_{i,d}\}.$$

This definition follows the extreme-value literature’s classical block maxima approach (Leadbetter et al., 2012; Coles et al., 2001). Because daily return series exhibit jumps, volatility clustering, and heavy-tailed distributions, the monthly MAX serves as a summary statistic capturing the most extreme upward shock within each monthly block.

A natural concern mirrors those raised in the realized-variance literature: the tradeoff between sampling frequency and measurement noise. While volatility forecasting uses high-frequency (5-minute or 1-minute) returns to estimate realized variance (e.g., Andersen et al., 2001; Bandi and Russell, 2006), extreme-return forecasting faces a different challenge. The continuous-time limit of the maximum depends not only on sampling frequency but also on the underlying jump process. In principle, intraday maxima could refine estimates of extreme behavior. However, as in the IVOL forecasting setting, practical constraints force us to rely on daily returns because high-frequency data for the majority of U.S. firms are only available since the early 1990s and severely restricted for delisted or low-priced stocks.

Daily block maxima are therefore used throughout our sample (1962–2023), consistent with the data availability constraints identified in the realized-variance literature (Han and Lesmond, 2011). While daily sampling may understate the true continuous-time maximum within a month, the bias is systematic and affects all observations symmetrically across firms and over time.

Because MAX is constructed as an extreme order statistic, its distribution exhibits important properties: **Non-Gaussianity**: the distribution is heavy-tailed and right-skewed, even if returns themselves are symmetric. **Jump sensitivity**: MAX responds disproportionately to rare, high-magnitude price movements (e.g., earnings announcements, takeover rumors, litigation outcomes). **High variance and measurement noise**: MAX contains more sampling noise than sample means or volatilities (consistent with the extremal-sample literature).

Consequently, any predictive model for MAX must account for the fact that MAX reflects tail behavior rather than central-moment dynamics.

3.4 Predictive Models for Future MAX

We now introduce four primary forecasting models. Our modeling approach parallels the structure of the realized-variance forecasting literature, which compares martingale benchmarks, ARMA specifications, mixed-frequency models, and nonlinear extreme-value models (Bandi and Russell, 2006; Ghysels et al., 2019). We show that each of our four models corresponds to a distinct theoretical motivation.

Let $E_{\text{MAX},i,t+1}^{(N)}$ denote the expected MAX for firm i in month $t + 1$ produced by forecasting model N , where $N \in \{1,2,3,4\}$.

3.4.1 AR(1) Model for Monthly MAX

The simplest baseline model assumes that monthly maxima follow a first-order autoregressive process:

$$\text{MAX}_{i,t+1} = a_i + \phi_i \text{MAX}_{i,t} + \varepsilon_{i,t+1}.$$

This specification is motivated by the persistence of volatility and tail risk documented in the empirical literature. As volatility clusters and jump intensities evolve smoothly over time, the magnitude of recent maxima carries forward information about the expected severity of future tail events. This logic mirrors the use of AR(1) models for realized variances (French, Schwert, and Stambaugh 1987; Bollerslev et al., 2016), except applied to the extreme tail rather than the second moment.

We estimate the AR(1) parameters using an expanding window of historical MAX values, requiring a minimum of 120 non-missing month-to-month pairs $(\text{MAX}_{i,t}, \text{MAX}_{i,t+1})$. The forecast is the standard AR(1) one-step-ahead projection:

$$E_{\text{MAX},i,t+1}^{(1)} = \hat{a}_{i,t} + \hat{\phi}_{i,t} \text{MAX}_{i,t}.$$

This parsimonious predictor provides a baseline against which more sophisticated models can be compared.

3.4.2 ARMA(1,1) Model: Measurement Noise and Jump Shocks

While the AR(1) model is a natural starting point, it is almost surely misspecified. Analogous to the results in Andersen et al. (2003) for realized variances, the maximum of a noisy high-frequency process with persistent volatility and microstructure-like noise naturally leads to an ARMA(1,1) representation. To see this, suppose daily returns follow:

$$r_{i,d} = \sigma_{i,d} z_{i,d}, z_{i,d} \sim \text{i.i.d.},$$

with persistent volatility $\sigma_{i,d}$. Then, the monthly maximum depends on extreme observations drawn from this time-varying distribution. As shown in the realized-variance literature, when the underlying latent process is persistent and observed with noise, the empirically measured block statistic inherits an MA(1) structure (Bandi and Russell, 2006; Andersen et al., 2003).

We therefore estimate:

$$\text{MAX}_{i,t} = \mu_i + \phi_i \text{MAX}_{i,t-1} + \theta_i \varepsilon_{i,t-1} + \varepsilon_{i,t},$$

where θ_i captures short-lived shocks that quickly decay. Economically, a large extreme return in one month may partially reflect transient events (e.g., earnings surprises) that do not persist into the next month; the MA(1) term allows the model to counteract persistent AR-driven forecasts following abnormally large maxima.

The ARMA(1,1) parameters are estimated monthly using a rolling window of up to 120 months, with refitting every 12 months to manage computational costs. The forecast is:

$$E_{\text{MAX},i,t+1}^{(2)} = \hat{\mu}_{i,t} + \hat{\phi}_{i,t} \text{MAX}_{i,t} + \hat{\theta}_{i,t} \hat{\varepsilon}_{i,t}.$$

We expect this model to outperform the AR(1) specification when extreme movements are accompanied by transitory jumps or measurement noise, analogous to the realized-variance setting.

3.4.3 Quantile–Tail Approximation Model

Extreme-value theory provides analytical results describing the behavior of maxima in Gaussian or near-Gaussian settings (Leadbetter et al., 2012; Embrechts et al., 1997). In particular, the expected maximum of n independent draws from a distribution can be approximated using high-quantile expansions of the form:

$$E[\max(r_1, \dots, r_n)] \approx Q^{(p)} + \sigma(z_n - z_p),$$

where $Q^{(p)}$ is a tail quantile (99th percentile), σ is the scale of dispersion, z_p is the corresponding normal-quantile, and z_n approximates the expected extreme quantile of n draws.

We adopt this quantile-tail formula by computing: rolling 252-day 99th percentiles $Q_{i,t}^{(0.99)}$, rolling standard deviations $\sigma_{i,t}$, the expected next-month draw count $n_{i,t+1}$, the quantile gap $z_{n_{i,t+1}} - z_{0.99}$.

The forecast is then:

$$E_{\text{MAX},i,t+1}^{(3)} = Q_{i,t}^{(0.99)} + \sigma_{i,t}(z_{n_{i,t+1}} - z_{0.99}).$$

This method captures tail behavior without assuming a parametric time-series model for monthly MAX. It is a nonparametric approximation grounded in the asymptotic distribution of order statistics.

3.4.4 EVT–GPD Peaks-over-Threshold Model

The fourth model draws directly from classical extreme-value theory. Following Pickands (1975) and Balkema and de Haan (1974), exceedances above a high threshold u converge in distribution to the Generalized Pareto Distribution:

$$(r - u \mid r > u) \sim \text{GPD}(\xi, \sigma).$$

The MAX of a block of observations can be expressed in terms of the tail distribution governing exceedances. Accordingly, we estimate the GPD parameters $(\xi_{i,t}, \sigma_{i,t})$ using the exceedances above the rolling 95th percentile of daily returns over a 500-day window. Let $p_{u,i,t}$ denote the empirical exceedance probability and $m_{i,t} = n_{\text{next}}p_{u,i,t}$ the expected number of exceedances in the next month.

The EVT-based expected maximum is:

If $\xi \approx 0$ (Gumbel case):

$$E_{\text{MAX},i,t+1}^{(4)} = u_{i,t} + \sigma_{i,t} \log(m_{i,t}).$$

If $\xi \neq 0$:

$$E_{\text{MAX},i,t+1}^{(4)} = u_{i,t} + \frac{\sigma_{i,t}}{\xi_{i,t}} (m_{i,t}^{\xi_{i,t}} - 1).$$

This model is theoretically the most aligned with the probabilistic behavior of extreme returns (Embrechts et al., 1997). It directly models the shape and scale of the return distribution's tail rather than relying on autoregressive dynamics or Gaussian approximations.

3.5 Forecast Evaluation Framework

To assess the predictive content of the four expected-MAX models, we implement a comprehensive out-of-sample forecast evaluation procedure consistent with the forecast-comparison literature (Diebold and Mariano, 2002; West, 2006). For each firm, we align all forecasts, AR(1), ARMA(1,1), Quantile–Tail, EVT–GPD, and a martingale benchmark on a common set of month-end observations where the realized next-month maximum return $\text{MAX}_{i,t+1}$ is available. All forecasts are computed strictly out of sample, using only information observable at the end of month t . To prevent a small number of extreme estimates from dominating the evaluation, we clip forecasted MAX values to a plausible range $[0, 0.5]$, a transformation that preserves relative performance rankings but mitigates numerical instability.

The first part of the evaluation computes firm-level RMSE, the standard measure of forecast accuracy in the volatility and tail-risk literature (Andersen et al., 2003; Ghysels et al., 2019). For each firm i and model m , we compute

$$\text{RMSE}_i^{(m)} = \sqrt{\frac{1}{T_i} \sum_{t=1}^{T_i} \left(\text{MAX}_{i,t+1} - \widehat{\text{MAX}}_{i,t+1}^{(m)} \right)^2},$$

requiring at least 24 out-of-sample months per firm to ensure stable estimates. This provides a model-by-model distribution of forecast errors across firms and serves as the basis for all subsequent comparisons.

To examine whether differences in accuracy are systematic, we evaluate pairwise RMSE differences across models. For each pair (m_1, m_2) , we compute the firm-level differences $\text{RMSE}_i^{(m_1)} - \text{RMSE}_i^{(m_2)}$ and apply three complementary statistical tests. First, a one-sample t-test evaluates whether the mean RMSE difference is significantly different from zero, capturing average improvements in accuracy. Second, a sign test assesses whether one model outperforms another for a majority of firms, focusing solely on the direction of improvement. Third, a Wilcoxon signed-rank test evaluates whether the distribution of RMSE differences differs from zero without relying on normality assumptions, which is important for heavy-tailed tail-risk forecasts. This multi-test structure mirrors common practice in comparative forecast-evaluation settings in empirical finance (e.g., Bollerslev et al., 2016).

Finally, we analyze distributional properties and dependence across forecasts. For each model, we compute the cross-sectional percentiles of the percentage performance loss relative to the firm-specific best model, which highlights how far a model typically is from the most accurate attainable forecast for a given firm. We also compute forecast correlations by estimating, for each firm, the correlation matrix of the forecast series across models and then averaging these matrices. This reveals whether certain approaches produce highly similar predictions or whether they capture distinct components of extreme-return dynamics, analogously to the way MIDAS, quarticity-adjusted, and ARMA-based volatility models often exhibit different dependence structures.

3.6 Expected and Unexpected Components of Maximum Daily Returns

A central objective of this paper is to separate the portion of next-month maximum daily returns that is predictable from information available at time t , and the portion that represents a genuine unexpected shock. To do this, we construct two related objects: (i) the expected surprise $D_{i,t}$, which reflects how next month's predicted MAX compares to this month's realized MAX, and (ii) the unexpected MAX $UMAX_{i,t+1}$, which is the forecast error realized in month $t + 1$.

(a) Expected Surprise at Time t

At the end of month t , before month $t + 1$ has realized, our forecasting models generate an expectation of next month's maximum daily return, denoted

$$E_{MAX,i,t+1|t}.$$

We compare this forecast to the current realized maximum daily return in month t , $MAX_{i,t}$, by computing the expected surprise

$$D_{i,t} = MAX_{i,t} - E_{MAX,i,t+1|t}.$$

The sign of $D_{i,t}$ has a clear economic interpretation. If $D_{i,t} < 0$ then,

$$E_{MAX,i,t+1|t} > MAX_{i,t},$$

meaning the model predicts that the stock will be more jumpy or experience larger daily extremes in the upcoming month than it did in the current month. In other words, the stock is expected to face higher future tail intensity.

If $D_{i,t} > 0$ then,

$$E_{MAX,i,t+1|t} < MAX_{i,t},$$

meaning the model predicts a decline in jump intensity next month. The stock is expected to have quieter or less extreme daily movements.

It is important to emphasize that $D_{i,t}$ is not a statistical surprise, it is computed before month $t + 1$ realizes. Rather, it captures the model-implied expected direction and magnitude of change in maximum daily returns relative to the current month. Conceptually, $D_{i,t}$ is an anticipated trend in tail risk, not an innovation.

This quantity is used to classify stocks into two economically distinct regimes: those for which future jump intensity is expected to increase ($D_{i,t} < 0$) and those for which it is expected to decrease ($D_{i,t} > 0$). Later, in Section 3.6, this conditioning structure plays a central role in our portfolio analyses.

(b) Unexpected MAX (Realized Shock) at Time $t + 1$

Once month $t + 1$ has fully realized, we observe the actual maximum daily return, $MAX_{i,t+1}$. The unexpected MAX, or MAX shock, is then defined as

$$UMAX_{i,t+1} = MAX_{i,t+1} - E_{MAX,i,t+1|t}.$$

This object represents the portion of extreme return behavior that could not be forecasted from the information available at the end of month t . A large positive $UMAX$ indicates an unexpectedly large jump in returns, whereas a negative value indicates a quieter-than-expected month. The terminology “unexpected MAX” follows directly from its interpretation as a forecast error or innovation in the dynamics of maximum daily returns.

Economically, $UMAX_{i,t+1}$ captures unanticipated tail realizations, which may arise from firm-specific news, earnings-related volatility bursts, liquidity shocks, or market-wide jump events. Because it isolates the genuinely unpredictable component of extreme returns, $UMAX$ is especially useful for examining whether the MAX anomaly is driven by expected jumpiness or by unexpected shocks that are priced in the cross-section.

3.7 Portfolio Sorting Procedures and Fama–MacBeth Regressions

To evaluate whether expected and unexpected components of maximum daily returns carry priced information, we employ a series of portfolio-sorting procedures and Fama–MacBeth regressions. These approaches provide complementary nonparametric and parametric perspectives on how MAX-related characteristics map into next-month returns, following the standard empirical design of the asset-pricing literature (e.g., Jegadeesh and Titman, 1993; Fama and French, 2008).

We begin with univariate portfolio sorts. At the end of each month t , stocks are independently sorted into five quintile portfolios based on a single variable. We conduct such sorts for realized maximum daily return (the martingale benchmark), for each of the four model-based forecasts $E_{MAX}^{(N)}$, and for each measure of unexpected MAX $UMAX^{(N)}$. Because the expected surprise

$$D_{i,t} = MAX_{i,t} - E_{MAX,i,t+1|t}$$

classifies stocks into contrasting informational environments, those expected to experience higher future jump intensity ($D_{i,t} < 0$) and those expected to experience lower future jump intensity ($D_{i,t} > 0$), we conduct all quintile sorts separately within these two subsamples. This conditioning structure allows us to examine whether the pricing of MAX signals differs depending on whether the month-ahead jumpiness is expected to rise or fall. After forming quintile portfolios, we compute their equal-weighted and value-weighted returns in month $t + 1$, which provide a nonparametric mapping between MAX-related characteristics and future stock performance.

To further disentangle the roles of predictable and unpredictable components of extreme returns, we also perform bivariate (double) sorts. At the end of each month t , we first sort firms into quintiles on one dimension, such as unexpected shocks $UMAX^{(N)}$, realized MAX, or expected MAX, and then, within each quintile, we sort firms again into quintiles based on a second dimension. This procedure results in a 5×5 grid of 25 portfolios for each pair of characteristics. Typical combinations include sorting on $UMAX$ first and then sorting on realized MAX, or sorting on expected MAX first and then sorting on the magnitude or direction of the unexpected component. These 25-portfolio grids allow us to capture how future returns vary with the interaction of expected jumpiness, realized jump intensity, and unanticipated tail shocks. This approach parallels two-dimensional sorts used to distinguish between risk-based and behavioral mechanisms in the cross-section of volatility, skewness, and tail-risk characteristics.

As a parametric complement to these portfolio sorts, we estimate Fama–MacBeth (1973) cross-sectional regressions. Each month, we regress next-month returns on MAX-related characteristics observed at the end of month t . We begin with univariate specifications, regressing returns on

realized MAX, each expected MAX forecast, and each unexpected MAX shock separately, which allows for a clean estimate of the marginal price of each characteristic. We then estimate a multivariate specification of the form:

$$R_{i,t+1} = \alpha_t + \beta_t E_{\text{MAX},i,t+1}^{(N)} + \gamma_t UMAX_{i,t+1}^{(N)} + \delta_t^\top Z_{i,t} + \varepsilon_{i,t+1},$$

where $Z_{i,t}$ includes standard firm-level controls used in cross-sectional asset-pricing tests: market beta, size, book-to-market ratio, momentum, short-term reversal, illiquidity (Amihud measure), idiosyncratic volatility, and idiosyncratic skewness. Averaging coefficient estimates across months yields inference on whether expected extreme return behavior, unexpected jump shocks, or both are systematically priced.

4. Empirical Results

This section presents the main empirical findings of the paper. We begin by evaluating the ability of different models to forecast future extreme returns and then examine the implications of these forecasts for the cross-section of stock returns. The objective is to determine whether predictable variation in extreme return realizations contains information about future returns and whether separating predictable and unpredictable components helps clarify the economic mechanism behind the MAX anomaly.

We first assess the out-of-sample forecasting performance of the proposed models for predicting the maximum daily return within the following month. This step is important because the empirical analysis relies on the accuracy of these forecasts to construct measures of expected and unexpected extreme returns. By comparing several forecasting approaches, we evaluate whether extreme return behavior exhibits systematic predictability that can be captured using historical information.

After assessing the models' forecasting performance, we investigate the cross-sectional return implications of the forecasts. In particular, we examine whether expected extreme returns are related to future stock returns and whether this relationship differs from the one obtained using realized maximum returns. We also explore how unexpected shocks to extreme returns affect subsequent performance and whether these shocks help explain the observed return patterns associated with lottery-like stocks.

4.1. Forecast Performance

4.1.1 Forecasting Setup

The forecasting exercise follows the structure of Andersen et al. (2003), Bekaert et al. (2025), and related volatility-forecasting work, adapted here to the monthly maximum daily return (MAX). All models are estimated using rolling windows of past information, and each specification generates a forecast of next month's maximum daily return, $EMAX_{i,t+1|t}$. To be included in the forecasting sample, a firm must have at least 36 monthly observations. While we conduct full in-

sample estimation for completeness, the discussion below focuses on the out-of-sample results, as both approaches yield qualitatively similar conclusions.

At each month-end t , we estimate model parameters using the last ten years of monthly MAX observations (or all available data for younger firms). This rolling-window design balances estimation efficiency with flexibility to accommodate changes in the tail behavior of firm-level returns. For each firm-month, we produce an out-of-sample forecast for month $t + 1$, and consistent with the practice in high-frequency volatility forecasting, we winsorize extreme forecasts at economically reasonable bounds to remove unrealistic outliers. We evaluate forecasting accuracy across models using the root mean squared error (RMSE), our primary metric, and also compute: (i) the fraction of firm-months for which a given model delivers the lowest RMSE, (ii) the average rank across models, and (iii) the “improvement-to-worst” statistic comparing each model to the worst performer. Finally, to formally assess whether differences in RMSEs are statistically significant, we perform three standard paired tests: the Student t-test, the Sign test, and the Wilcoxon signed-rank test.

The five forecasting models evaluated are: (1) a naïve martingale, (2) an AR(1) specification, (3) an ARMA(1,1) model, (4) a Quantile/Tail-proxy model based on daily extreme-quantile extrapolation, and (5) an EVT–GPD model relying on tail exceedances.

Together, they span persistence-based, distribution-based, and extreme-value frameworks.

4.1.2 Out-of-Sample Performance of Base Models

Table 1 summarizes the out-of-sample forecasting performance. The martingale model produces the weakest forecasts, with the highest mean RMSE (0.0629), the worst average rank (4.43), and the lowest improvement-to-worst statistic (15.85%). In contrast, the ARMA(1,1) model is clearly the dominant specification. It achieves the lowest mean RMSE (0.0535), the lowest median RMSE (0.0378), the best average rank (1.82), and is the best model in 46.27% of all firm-months, more than twice as often as any other model. The AR(1) model performs second best, with a mean RMSE of 0.0548 and an average rank of 2.45, followed by the EVT–GPD model and the Quantile/Tail proxy, which achieve intermediate levels of accuracy.

The improvement-to-worst measure reinforces these results. The ARMA(1,1) model captures nearly 88% of the maximum attainable improvement relative to the worst model, substantially outperforming the AR(1) model (78.70%) and the EVT–GPD model (64.33%). The tail-quantile proxy performs better than the martingale but worse than the autoregressive models, consistent with tail-quantile adjustments capturing some, but not all, extreme-return dynamics.

The evidence supports the ARMA(1,1) model as the most accurate and robust predictor of future MAX values, consistent with the long-standing result that ARMA models efficiently capture short-term persistence in realized extreme outcomes.

4.1.3 Formal Pairwise Tests of Forecasting Differences

Table 2 reports the paired significance tests across all model combinations. Given the large number of firm-month observations, all three tests exhibit substantial statistical power. Across Panels A–C, the ARMA(1,1) model produces significantly lower RMSEs than every other model, with t-statistics exceeding ± 29 in all comparisons and p-values effectively equal to zero. The Sign and Signed-Rank tests confirm this dominance, demonstrating that the ARMA model systematically outperforms alternatives in both the direction and magnitude of forecast improvements.

The AR(1) model is consistently the second-best performer, producing significantly lower RMSEs than the Quantile/Tail and EVT–GPD models. The nonparametric tests particularly emphasize the gap: for example, AR(1) versus EVT–GPD yields Sign-test statistics of -36.17 and Signed-Rank statistics of 16.91 (all p-values = 0.00). Meanwhile, the martingale model is significantly worse than every other specification, reflecting its inability to account for persistence or tail dynamics in MAX.

4.2 Cross-Model Performance Distribution and Forecast Correlation

Table 3 provides insight into cross-sectional variation in forecast accuracy. Panel A compares each model to the “best” model for each firm-month. The martingale model shows large deviations: its RMSE is 36–50% worse than the best model for firms at the upper quartiles. In sharp contrast, the ARMA(1,1) model remains extremely close to the best model even when it is not the winner; for 50% of firms, its RMSE is less than 0.38% above the best, and even at the 90th percentile it is only 10.58% worse. The AR(1) model also exhibits strong robustness, with median deviations of just 3.41%. The EVT–GPD and tail-quantile models provide moderate accuracy but larger deviations in the upper percentiles.

Panel B shows correlations among forecasts. All models correlate positively, but the strength of the correlations reflects differences in modeling philosophy. Persistence-based models (Martingale, AR(1), ARMA(1,1)) are highly correlated with each other (0.63–0.90), whereas correlations between autoregressive and tail-based models are weaker (0.25–0.57). The EVT–GPD and Quantile/Tail proxy forecasts correlate strongly with each other (0.62), indicating that they capture related features of tail behavior. The martingale forecast exhibits the lowest correlations with the tail-adjusted models, consistent with its lack of distributional structure.

These results underscore two key points. First, the ARMA(1,1) model provides not only the most accurate forecasts on average but also highly stable performance across the cross-section. Second, although tail-based methods incorporate important information about extreme return events, they cannot fully substitute for the predictive content embedded in autoregressive dynamics.

4.3 Expected MAX and Return Predictability

We begin by examining whether expected extreme return behavior predicts future stock returns and whether it reproduces the negative return patterns traditionally associated with realized

maximum daily returns. At the end of each month, stocks are sorted into quintile portfolios based on realized MAX and, separately, based on each model-based forecast of expected MAX. Table 4 reports next-month portfolio returns under both equal-weighted and value-weighted schemes, together with corresponding factor-adjusted alphas.

Sorting on realized MAX reproduces the MAX anomaly in a clear and economically meaningful manner. Under equal weighting, average next-month returns decline monotonically from 1.25% in the lowest realized-MAX quintile to 0.71% in the highest quintile, yielding a high-minus-low return spread of -0.55% per month with a t-statistic of -2.40 . This negative spread remains strong after adjusting for common risk factors: the FF6-adjusted alpha of the high-minus-low portfolio is -0.83% per month ($t = -3.13$), and the q-factor alpha is -0.86% ($t = -3.36$). The value-weighted results reinforce this pattern. Returns fall from 1.00% in the lowest quintile to 0.28% in the highest, producing a spread of -0.72% ($t = -2.67$), with factor-adjusted alphas close to -1% per month. These magnitudes confirm that stocks experiencing unusually large daily price movements in the recent past earn significantly lower subsequent returns.

In contrast, portfolios formed on expected MAX display a fundamentally different return pattern. Across all four forecasting models, expected returns do not decline across expected-MAX quintiles. Instead, under equal weighting, returns are generally flat or increasing. For example, when sorting on ARMA-based expected MAX, average returns rise from 1.11% in the lowest quintile to 1.33% in the highest, producing a positive high-minus-low spread of 0.23%. Similar patterns emerge for the other models. The AR(1)-based expected MAX yields a spread of 0.25%, the Quantile/Tail proxy yields 0.21%, and the EVT–GPD forecast yields 0.15%.

The factor-adjusted results reinforce this conclusion. Unlike realized MAX, sorting on expected MAX does not generate large negative alphas. Under equal weighting, the FF6-adjusted high-minus-low alphas range between -0.12% and -0.31% per month, with t-statistics well below conventional significance thresholds. The q-factor alphas are similarly small in magnitude and statistically insignificant. Under value weighting, expected-MAX spreads are slightly negative in some specifications, for instance, -0.23% for ARMA-based expected MAX, but remain economically modest and statistically weak compared to the realized-MAX benchmark.

The contrast between realized and expected MAX is economically revealing. Realized MAX captures extreme price movements that have already occurred, which may reflect transient price pressure, investor overreaction, or episodic shocks. Expected MAX, by construction, isolates the predictable component of future jumpiness based on historical information. The absence of a negative return premium for expected MAX, even when using models specifically designed to capture tail behavior, indicates that predictable exposure to future extreme returns does not drive the MAX anomaly.

Table 5 confirms these portfolio-level findings in a parametric setting using univariate Fama–MacBeth cross-sectional regressions. When next-month returns are regressed on realized MAX, the estimated slope coefficient is -3.49 with a t-statistic of -3.66 , indicating a strong and statistically significant negative relation between realized extreme returns and subsequent

performance. In contrast, when returns are regressed on expected MAX, the coefficients are uniformly small and statistically insignificant across all forecasting models. For example, the coefficient on ARMA-based expected MAX is 0.82 ($t = 0.32$), while the EVT-GPD-based expected MAX yields 0.86 ($t = 0.24$). Although the adjusted R^2 values are slightly higher for expected-MAX regressions than for the martingale benchmark, they remain modest and do not suggest meaningful standalone return predictability.

While realized MAX is associated with large and statistically significant negative future returns, expected MAX is not. Predictable variation in future extreme return behavior does not generate the MAX anomaly. This result implies that the negative return premium associated with extreme price movements must arise from components of realized MAX that are not anticipated *ex ante*. In the next subsection, we investigate this hypothesis directly by conditioning on expected changes in MAX and by explicitly decomposing realized MAX into expected and unexpected components.

4.4 Expected versus Unexpected MAX and Return Predictability

This section decomposes the MAX effect into its expected and unexpected components. Using the forecasts developed in Section 3, we distinguish between (i) variation in MAX that is anticipated at time t , and (ii) shocks to MAX that materialize unexpectedly at $t + 1$. The objective is to identify whether the negative return predictability traditionally associated with high MAX is driven by *ex-ante* expectations about future extreme returns, or by *ex-post* realizations of extreme outcomes.

4.4.1 Expected changes in MAX: portfolio evidence based on $D_{i,t}$

We begin by examining return predictability associated with the expected change in MAX, defined as

$$D_{i,t} = MAX_{i,t} - E_MAX_{i,t+1|t}.$$

Negative values of $D_{i,t}$ indicate that the model predicts an increase in MAX in the subsequent month relative to the current month, while positive values indicate an expected decline in future MAX. We therefore partition stocks into two mutually exclusive sets based on the sign of $D_{i,t}$ and, within each set, form quintile portfolios based on the magnitude of $D_{i,t}$.

Panel A of Table 6 reports equal-weighted portfolio returns for stocks with $D_{i,t} > 0$, i.e., firms for which MAX is expected to decline. Across all forecasting models, returns decline monotonically from low to high $D_{i,t}$. For example, under the EVT specification, average monthly returns fall from 1.29% in the lowest quintile to 0.66% in the highest quintile, yielding a high-minus-low spread of -0.63% with a t -statistic of -3.50 . Similar, though slightly weaker, patterns appear for the ARMA and quantile-based forecasts. Risk-adjusted alphas are economically large and statistically significant. These results indicate that when investors expect MAX to fall sharply, stocks with the largest expected declines in jump risk subsequently earn the lowest returns.

Panel B of Table 6 considers the complementary case in which $D_{i,t} < 0$, meaning that MAX is expected to rise. In contrast, portfolio returns now increase monotonically with the magnitude of the expected increase in MAX. Under the EVT forecast, returns rise from 1.29% in the lowest quintile to 1.64% in the highest quintile, producing a positive spread of 0.35% ($t = 2.10$). Importantly, however, these return differentials are largely absorbed by standard risk factors. This asymmetry suggests that expected increases in MAX are not penalized in equilibrium, whereas expected decreases are associated with economically meaningful underperformance.

Value-weighted results, reported in Table 6B (Appendix A), reinforce these conclusions. While the magnitudes of the spreads are attenuated, expected declines in MAX continue to generate significantly negative alphas, particularly under the EVT and quantile-based forecasts. The evidence indicates that expected variation in MAX matters for returns, but only asymmetrically, and does not by itself reproduce the classic MAX anomaly.

4.4.2 Unexpected MAX shocks: portfolio evidence based on *UMAX*

We next turn to the unexpected component of MAX, defined as

$$UMAX_{i,t+1} = MAX_{i,t+1} - E_{t-1}MAX_{i,t+1|t}.$$

Unlike $D_{i,t}$, which reflects anticipated changes in jumpiness, *UMAX* captures realized extreme returns that were not forecast at time t . If the MAX effect is driven by surprise realizations of extreme outcomes rather than by expectations, return predictability should be concentrated in this component.

Panel A of Table 7 presents equal-weighted portfolios sorted on *UMAX*. Across all forecasting methods, returns decline sharply as unexpected MAX increases. Using the ARMA forecast, average returns fall from 1.41% in the lowest quintile to 1.00% in the highest quintile, yielding a -0.41% spread with a t -statistic of -5.46 . The corresponding FF6 alpha is -0.83% ($t = -7.15$), and the q -factor alpha is -0.88% ($t = -7.01$). Similar or stronger patterns are observed for the quantile and EVT-based measures, with high-minus-low return spreads exceeding -0.50% per month and alphas approaching -1% per month.

Panels B and C further decompose these effects by conditioning on the sign of *UMAX*. Panel B focuses on stocks experiencing unexpectedly high MAX realizations ($UMAX > 0$). In this subsample, return spreads are particularly pronounced. Under the EVT measure, returns decline from 1.29% to 0.66%, producing a -0.63% spread ($t = -3.51$), with FF6 and q -factor alphas of -1.06% and -1.09% , respectively. In contrast, Panel C shows that when $UMAX < 0$, unexpectedly low MAX realizations, returns increase monotonically, but the corresponding alphas are small and statistically insignificant. Thus, the return penalty is concentrated entirely in positive surprise MAX realizations.

Value-weighted results in Table 7B (Appendix A) confirm the robustness of these findings. Although spreads are smaller, unexpectedly large MAX realizations continue to generate statistically significant negative alphas, particularly under the EVT-based measure. Across

specifications, the evidence consistently indicates that unexpected MAX shocks are the primary driver of the MAX anomaly.

The combined evidence from Tables 6 and 7 leads to three central conclusions. First, expected changes in MAX predict returns only asymmetrically. Stocks expected to experience sharp declines in jump risk earn significantly lower returns, whereas expected increases in MAX are not penalized. Second, unexpected MAX realizations exhibit strong, monotonic, and economically large return predictability, with negative alphas concentrated in stocks experiencing unexpectedly extreme returns. Third, once MAX is decomposed into expected and unexpected components, the traditional MAX effect is revealed to be almost entirely attributable to the unexpected component.

These findings mirror the decomposition results in the expected idiosyncratic volatility literature and suggest that the MAX anomaly is not compensation for ex-ante jump risk. Instead, it reflects investor reactions to unexpected extreme outcomes, consistent with behavioral explanations emphasizing lottery-like payoffs and surprise-driven mispricing.

4.5. Unexpected MAX, Extreme Return Realizations, and the Source of Return Predictability

The return predictability often attributed to lottery-like payoffs and extreme positive daily returns is not a generic “high MAX leads to low future return” relation. Instead, it is concentrated in states where extreme realizations arrive as surprises relative to what was predictable ex ante. The objective here is therefore diagnostic, not merely confirmatory. If the MAX anomaly were primarily a compensation for expected jump risk (a priced risk premium), then conditioning on expected MAX should preserve a stable relation between MAX exposure and subsequent returns. If, instead, the anomaly reflects mispricing triggered by unexpected extreme outcomes, for example, attention shocks, extrapolative belief updating, or lottery-demand episodes, then return predictability should be strongest precisely when the realized extreme event is unexpectedly large. To distinguish between these narratives, we use two complementary double-sort designs that jointly condition on (i) the unexpected component of MAX and realized MAX, and (ii) the unexpected component of MAX and expected MAX.

We begin from the decomposition of next month’s realized MAX into an expected component and an innovation. For each forecasting model m (AR, ARMA, QTAIL, EVT), we construct expected MAX at time t , denoted $E_MAX_{i,t+1|t}^{(m)}$, and define the unexpected component as

$$U_MAX_{i,t+1}^{(m)} \equiv MAX_{i,t+1} - E_MAX_{i,t+1|t}^{(m)}.$$

We label this component “unexpected” because it is, by construction, the forecast error that is part of the realized extreme return in $t + 1$ that cannot be inferred from the information used by the forecasting model at time t . Positive values ($U_MAX > 0$) correspond to negative surprise states, realized extremes are larger than expected. Negative values ($U_MAX < 0$) correspond to positive surprise states; realized extremes are milder than expected. This sign convention is economically

meaningful for our setting. A large realized MAX is not necessarily informative unless it is large relative to what was anticipated. The same realized MAX can be “routine” for a stock that is persistently jumpy, or “salient” for a stock for which such an extreme is unexpected. The double sorts operationalize exactly this distinction.

The key logic of the tests is simple. If the MAX effect is mainly about expected exposure to extreme payoff risk, then sorting on MAX (or on E_MAX) should predict returns regardless of whether U_MAX is positive or negative. Under a mispricing mechanism, however, the anomaly should be state dependent: unexpectedly large extremes ($U_MAX > 0$) should trigger the strongest subsequent underperformance, consistent with transient overvaluation after attention-grabbing events. Conversely, when realized extremes are smaller than expected, there is no “shock” to attention or beliefs, and therefore the subsequent return pattern should weaken or disappear.

4.5.1 $U_MAX \times MAX$

We first implement the joint test using realized MAX. Table 8 reports the equal-weighted double sorts $U_MAX \times MAX$ (shown for the ARMA-based implementation in the table presented here), split into two panels by the sign of unexpected MAX: Panel A conditions on $U_MAX > 0$ and Panel B conditions on $U_MAX < 0$. Appendix A3 reports the value-weighted counterpart. The design is 5×5 : each month we form quintile portfolios on unexpected MAX and quintile portfolios on realized MAX within each surprise regime, yielding 25 portfolios per panel.

The results in Table 8, Panel A ($U_MAX > 0 \times MAX$) provide a sharp diagnostic signal. When extreme payoffs arrive as negative surprises ($U_MAX > 0$), the future returns of high-MAX stocks collapse relative to low-MAX stocks, and the magnitude grows as we move from low unexpected MAX to high unexpected MAX. The high-UMAX/high-MAX corner portfolio earns an average excess return spread (high minus low along the MAX dimension) of about -0.70% per month with $t = -2.34$. Risk adjustment strengthens the conclusion: the corresponding FF6 alpha spread is about -1.28% ($t = -4.02$) and the Q-factor alpha spread is about -1.30% ($t = -3.34$). These alphas are economically large by asset-pricing standards and statistically tight. In other words, conditional on the event being an unexpectedly large extreme return, stocks in the highest MAX quintile deliver strongly negative abnormal performance.

Equally important is the shape of the predictability. Within the $U_MAX > 0$ regime, returns decline as MAX rises, and the decline steepens in higher-UMAX quintiles. This interaction is the economic core: the MAX effect is not merely “high MAX underperforms,” but rather “high MAX underperforms when it is unexpected.” This pattern is exactly what one would expect if unexpectedly extreme realizations act as salience shocks. Such shocks increase investor attention, fuel lottery demand, and intensify extrapolative beliefs, all of which can temporarily inflate prices and generate subsequent negative abnormal returns as prices mean revert.

In contrast, Table 8, Panel B ($U_MAX < 0 \times MAX$) shows that the MAX effect largely disappears when extreme outcomes are less than expected. Along the MAX dimension, the return spread is small and statistically weak: the high-minus-low spread in average excess returns is about $+0.11\%$

with $t = 0.40$, and the corresponding FF6 and Q-factor alpha spreads are near zero and insignificant (approximately -0.35% with $t = -1.16$ for FF6 and -0.53% with $t = -1.76$ for QF in the high-minus-low comparisons shown). This asymmetry is crucial. When $U_MAX < 0$ realized extremes fail to provide a “shock” relative to expectations, there is no systematic reversal. The same realized MAX level does not carry the same meaning across states; its informational and behavioral impact depends on whether it represents an unanticipated event.

The value-weighted evidence in Table 8B (Appendix A) reinforces the message and strengthens economic plausibility. Under $U_MAX > 0$, the negative spreads become even larger in the high-UMAX/high-MAX corner: the high-minus-low MAX spread reaches roughly -1.60% per month ($t = -5.09$), with FF6 and Q-factor alpha spreads around -2.23% ($t = -6.28$) and -2.40% ($t = -6.58$), respectively. This is informative for two reasons. First, it indicates that the phenomenon is not confined to microcaps; it remains strong when weighting toward larger firms. Second, it suggests that the mechanism is not a mechanical artifact of illiquidity or tiny-stock noise. If anything, the strongest reversals occur where the price impact of surprise extremes is economically meaningful and where reversal can be detected even after factor adjustment.

The $UMAX \times MAX$ results provide a direct bridge between forecasting and asset pricing. The forecasting models deliver an empirically disciplined notion of what an “expected” extreme return is for each stock at each point in time. The surprise decomposition then identifies the months when the market experiences unexpectedly large jumpiness. If the MAX anomaly were driven by a persistent risk exposure, the sign split should not matter: the pricing of risk would depend on exposure, not on whether realizations came in above or below forecast. The fact that predictability concentrates in $U_MAX > 0$ months supports a mispricing interpretation in which unexpected extremes are the trigger, consistent with attention- and belief-based channels emphasized in the lottery-demand literature.

At a deeper level, the asymmetry also resolves a conceptual puzzle that often complicates interpretations of MAX-type anomalies. A high realized MAX can mean either (i) genuinely high expected tail risk, which might command a premium if priced, or (ii) an outlier realization that investors overweight. Our decomposition separates these two possibilities and shows that the economically important return predictability is anchored in the second: the surprise component.

4.5.2 $UMAX \times EMAX$

We next ask whether the same conclusion holds when we control for expected tail risk directly by replacing realized MAX with expected MAX. This is an even stricter test of the “expected risk premium” story. If expected jumpiness is the fundamental driver, then portfolios sorted on E_MAX should display stable return patterns regardless of whether U_MAX is positive or negative. Conversely, if unexpected extremes drive mispricing, the conditioning on U_MAX should again be decisive.

Table 9 reports equal-weighted double sorts $U_MAX \times E_MAX$, again split by the sign of unexpected MAX (Panel A for $U_MAX > 0$ and Panel B for $U_MAX < 0$). Table 9B (Appendix

A) reports value-weighted results. The evidence is strikingly consistent with the $UMAX \times MAX$ findings. In Table 9, Panel A ($UMAX > 0 \times EMAX$), high expected MAX stocks underperform sharply when they also experience unexpectedly large extremes. The high-minus-low spread across expected MAX within high-UMAX is approximately -0.73% per month with $t = -2.31$, and the abnormal performance is even stronger: FF6 alpha spread around -1.21% ($t = -3.78$) and Q-factor alpha spread around -1.29% ($t = -3.36$). This indicates that even after conditioning on ex ante expected tail risk, the return reversal is concentrated when realized extremes arrive as negative surprises.

Meanwhile, Table 9, Panel B ($UMAX < 0 \times EMAX$) again shows weak or absent predictability. The spread across expected MAX is small, about $+0.02\%$ with $t = 0.07$, and factor-adjusted spreads remain statistically weak. This is a powerful result: expected MAX by itself does not generate robust return predictability once one conditions on whether the realized extreme component was unexpectedly large or unexpectedly small. Expected tail risk is not sufficient; what matters is the surprise realization.

The value-weighted analysis in Appendix A4 provides additional nuance. Under $U_MAX > 0$, the high-minus-low expected MAX spread becomes very large for returns (approximately -1.19% , $t = -5.03$) and strongly negative for alphas (FF6 about -1.50% , $t = -4.88$; QF about -2.03% , $t = -5.48$). Under $U_MAX < 0$, the sign and significance weaken, with spreads around -0.48% ($t = -2.16$) and alphas that are negative but materially smaller. This pattern is consistent with the paper's central mechanism: when extreme outcomes arrive as negative surprises, they generate the strongest subsequent reversals; when they do not, predictability is muted and unstable. The fact that the "negative surprise" state is where the signal concentrates is precisely what makes the interpretation coherent and compelling.

The apparent MAX anomaly is largely a conditional phenomenon that emerges when an extreme payoff arrives unexpectedly. The conditioning matters in both ways we test it. When we hold fixed realized MAX and vary UMAX, predictability strengthens in $U_MAX > 0$ states and collapses in $U_MAX < 0$ states. When we hold fixed expected MAX and vary UMAX, the same asymmetry appears. This is not what a stable risk-premium story would predict. A risk-premium interpretation would imply that exposure to expected tail risk, proxied by either MAX persistence or E_MAX , should command a consistent price of risk. Instead, the evidence points to a mechanism where the realized surprise extreme event is the catalyst for subsequent underperformance.

This mechanism aligns naturally with behavioral and limits-to-arbitrage arguments in the lottery-like stocks literature. Unexpectedly large extremes are precisely the events likely to generate disproportionate attention, media coverage, and retail interest; they can lead investors to extrapolate rare payoffs, overweight small probabilities, and temporarily bid up prices. Under this narrative, the reversal is not a compensation for bearing risk but a correction from state-dependent overpricing. The very strong negative alphas in the $U_MAX > 0 \times$ high-MAX (or high-EMAX) corner portfolios provide a particularly direct manifestation of this channel. These are the stocks that simultaneously (i) deliver a salient "jackpot-like" realization and (ii) are, ex ante, in the region

where extreme outcomes are relevant. They are exactly the cases where an attention-driven run-up is most likely, and where mean reversion should subsequently be strongest.

Finally, the split by sign is a structural implication of the decomposition. If U_MAX is truly capturing the innovation in extreme payoffs, then positive innovations should be the ones that create overreaction and reversal, while negative innovations should not systematically generate the same overpricing channel.

4.6. Cross-Sectional Regressions: Expected versus Unexpected MAX

This section provides formal cross-sectional evidence on the return implications of expected and unexpected extreme returns using Fama–MacBeth (1973) regressions. While portfolio sorts in Sections 4.2–4.4 offer nonparametric insights, cross-sectional regressions allow us to jointly condition on expected and unexpected components of MAX and to assess whether the predictive content of unexpected MAX survives controls for a rich set of firm characteristics. The results are reported in Table 10, which presents bivariate and Table 11 shows multivariate specifications.

4.6.1. Bivariate regressions: separating expectation from surprise

Table 10 reports Fama–MacBeth cross-sectional regressions examining the relation between next-month stock returns and realized, expected, and unexpected components of maximum daily returns. Each column corresponds to a different expected-MAX forecasting model, including the martingale benchmark, AR(1), ARMA(1,1), the quantile-based tail proxy, and the EVT/GPD model. The purpose of these regressions is to formally evaluate whether the predictive power of MAX arises from predictable exposure to extreme returns or from unexpected realizations of extreme price movements.

Panel A presents regressions that include both realized MAX and expected MAX. The coefficient on realized MAX remains negative and statistically significant across all specifications, consistent with the standard MAX anomaly documented by Bali et al. (2011). At the same time, the coefficient on expected MAX is positive and statistically significant in several specifications. This pattern indicates that once realized extreme returns are controlled for, higher expected extreme returns are associated with higher subsequent returns. Economically, this result suggests that predictable exposure to extreme price movements does not generate the negative return premium typically attributed to MAX.

Panel B reports regressions that include expected MAX and unexpected MAX simultaneously. The results reveal a striking pattern. Across all specifications, the coefficient on U-MAX is strongly negative and highly statistically significant, with t-statistics ranging from approximately -4.7 to -8.0 . In contrast, the coefficient on E-MAX becomes small, statistically insignificant, and unstable in sign once U-MAX is included. These results indicate that the cross-sectional return predictability associated with extreme returns is concentrated in the unexpected component rather than in predictable variation in extreme return intensity.

Panel C further clarifies this mechanism by including realized MAX together with U-MAX. Once unexpected extreme returns are controlled for, the coefficient on realized MAX becomes weak and statistically insignificant across most specifications, while the coefficient on U-MAX remains negative and significant.

These findings convey a clear message. Conditional on expectations, stocks that experience unexpectedly large extreme returns earn significantly lower subsequent returns. By contrast, variation in expected MAX, reflecting anticipated jumpiness or tail exposure, does not robustly predict the cross-section of returns once surprises are accounted for. In other words, the return predictability traditionally associated with MAX is not driven by exposure to expected tail risk but is instead concentrated in the unexpected component of extreme outcomes.

This bivariate evidence formalizes the intuition developed in Sections 4.3 and 4.4. There, we showed that the MAX effect is strongest when realized extreme returns exceed expectations and largely absent when extreme outcomes are anticipated. Panel A of Table 10 demonstrates that this insight survives a formal regression framework and is not an artifact of portfolio construction.

4.6.2. Multivariate regressions: a horse race with known predictors

Table 11 extends the analysis by introducing a comprehensive set of control variables commonly used in the asset pricing literature. In addition to E-MAX and U-MAX, the regressions include market beta, firm size, book-to-market ratio, momentum, short-term reversal, illiquidity, idiosyncratic volatility, and idiosyncratic skewness. These controls capture well-established risk characteristics and behavioral anomalies, including lottery-like preferences and volatility-related effects.

The inclusion of these controls serves two purposes. First, it tests whether unexpected MAX merely proxies for known predictors such as idiosyncratic volatility or skewness. Second, it evaluates whether the pricing impact of U-MAX is incremental to a broad set of firm characteristics.

The results reinforce and sharpen the conclusions from the bivariate analysis. U-MAX remains strongly negative and statistically significant in every specification, with t-statistics typically between -3.3 and -7.5 . The magnitude of the coefficient is economically meaningful, indicating that unexpected extreme returns are associated with substantial reductions in expected future returns even after controlling for standard risk and characteristic-based predictors.

By contrast, E-MAX remains weak and statistically insignificant across all multivariate specifications. In several cases, its coefficient is negative, but it never approaches conventional significance levels. This asymmetry between expected and unexpected components is central: once the surprise component of MAX is isolated, the expected component carries little independent pricing information.

The adjusted R^2 values in Panel B increase to approximately 5–6%, comparable to those reported in prominent cross-sectional studies that combine multiple firm characteristics. Importantly, this

improvement in explanatory power is driven primarily by U-MAX rather than E-MAX, underscoring the relevance of unexpected tail realizations for the cross-section of returns.

The regression evidence in Table 11 provides strong support for an interpretation of the MAX anomaly rooted in surprise rather than risk exposure. If MAX were priced as compensation for bearing expected tail risk, one would expect E-MAX to enter positively or negatively in a stable and significant manner once U-MAX is controlled for. Instead, the data show that only the unexpected component matters.

This pattern is difficult to reconcile with standard risk-based explanations. Expected tail risk is, by construction, known to investors at time t and should therefore be priced *ex ante* if it represents a systematic risk. The fact that E-MAX does not robustly price returns suggests that anticipated jumpiness does not command a premium or discount in equilibrium.

In contrast, unexpected extreme returns plausibly reflect belief errors, attention shocks, or extrapolative behavior. A large unexpected MAX realization may temporarily inflate investor expectations or attract speculative demand, leading to subsequent return reversals as beliefs correct. The strong negative coefficient on U-MAX is consistent with this mechanism: stocks that experience unexpectedly large extremes subsequently underperform as prices revert.

Crucially, the persistence of U-MAX in multivariate regressions indicates that this mechanism is distinct from existing volatility- or skewness-based explanations. U-MAX captures a dimension of extreme outcomes that is orthogonal to average volatility and higher-moment risk proxies, highlighting the importance of surprise in the tails rather than tail exposure.

A potential concern is that the UMAX effect may simply reflect the well-documented short-term reversal anomaly, since stocks experiencing unexpectedly large extreme returns may also exhibit elevated prior-month returns. To address this concern, we augment our baseline Fama-MacBeth specifications with an interaction term between UMAX and short-term reversal ($UMAX \times REVERSAL$). Across all four forecasting models and both bivariate and multivariate specifications, the coefficient on UMAX remains negative and statistically significant after including the interaction term, indicating that the UMAX effect is not subsumed by short-term reversal. The interaction term itself is positive in all specifications and statistically significant in only two of eight cases, suggesting that the two effects are largely independent and that high prior-month returns modestly attenuate rather than amplify the UMAX premium. These results confirm that unexpected extreme returns capture a distinct dimension of return predictability that is not explained by the reversal effect.

5. Conclusion

This paper investigates the economic mechanism underlying the MAX anomaly, the empirical finding that stocks experiencing extremely large positive daily returns tend to earn low subsequent returns. While previous studies have documented this pattern using realized measures of extreme returns, the existing literature has largely treated maximum daily returns as a single object without

distinguishing between predictable and unexpected components. This paper revisits the anomaly by separating realized extreme returns into expected and unexpected elements and examining their respective roles in explaining the cross-section of stock returns.

To implement this approach, we develop several forecasting models for next-month maximum daily returns, including AR(1), ARMA(1,1), quantile–tail approximations, and an extreme-value-theory peaks-over-threshold model. Out-of-sample forecasting comparisons indicate that the ARMA(1,1) specification provides the most accurate predictions of future extreme returns. Using these forecasts, we decompose realized MAX into expected and unexpected components and analyze their implications for asset pricing using portfolio sorts and Fama–MacBeth regressions.

The empirical evidence delivers a consistent message. Expected MAX, which captures the predictable component of extreme return behavior, has little ability to explain the cross-section of stock returns once unexpected shocks are taken into account. In contrast, the unexpected component of extreme returns strongly predicts future performance. Stocks that experience unusually large extreme returns relative to what was anticipated subsequently earn significantly lower returns. This result holds across multiple forecasting models, alternative portfolio constructions, and risk-adjustment frameworks including the Fama–French six-factor model and the q-factor model.

These findings provide a new perspective on the MAX anomaly. Rather than reflecting compensation for predictable exposure to tail risk, the anomaly appears to be driven primarily by unexpected extreme realizations. When extreme positive returns arrive as surprises, they are associated with temporary price pressure and subsequent return reversals. This interpretation is consistent with the view that extreme positive returns attract short-lived demand from speculative investors, pushing prices away from fundamental values before they gradually revert.

The results therefore suggest that the cross-sectional predictability associated with MAX should be interpreted as a mispricing phenomenon rather than a reward for bearing systematic tail risk. By separating expectations from surprises in extreme return realizations, this paper clarifies the economic forces behind one of the most prominent anomalies related to lottery-like stocks.

Several directions remain for future research. First, further work could explore the behavioral mechanisms underlying the market reaction to unexpected extreme returns, particularly the role of retail trading, investor attention, and speculative demand. Second, future studies could investigate whether similar expectation-versus-surprise decompositions help explain other anomalies related to tail events, such as skewness, crash risk, or jump risk. Third, extending the analysis to international markets or alternative asset classes may help determine whether the role of unexpected extremes in driving return reversals is a general feature of financial markets. These avenues may further improve our understanding of how extreme outcomes influence asset prices and investor behavior.

6. References

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Table 1. Out-of-Sample Forecast Performance of Expected MAX Models

This table reports the out-of-sample forecast performance of alternative models for predicting next-month maximum daily stock returns (MAX). The sample includes all common stocks listed on NYSE, Amex, and Nasdaq from July 1962 to December 2023 with sufficient return history. Five forecasting models are evaluated: a martingale benchmark (random walk), an AR(1) model, an ARMA(1,1) model, a quantile–tail approximation (QTAIL), and an extreme-value-theory-based Peaks-over-Threshold model using the Generalized Pareto Distribution (EVT/GPD). Forecast accuracy is measured using firm-level root mean squared error (RMSE), computed strictly out of sample. “Mean” and “Median” report the cross-sectional average and median RMSE across firms. “Best” reports the percentage of firms for which a given model delivers the lowest RMSE among all models. “Rank” reports the average rank of each model’s RMSE across firms. “Improve to Worst” reports the average percentage reduction in RMSE relative to the worst-performing model for each firm.

Forecast Model	Mean	Best	Rank	Improve to Worst	Median
Martingale	0.0629	3.86%	4.43	15.85%	0.0481
AR (1)	0.0548	23.20%	2.45	78.70%	0.0396
ARMA (1,1)	0.0535	46.27%	1.82	88.42%	0.0378
QTAIL	0.0566	16.09%	3.18	59.74%	0.0402
EVT/ GPD	0.0563	10.58%	3.12	64.33%	0.0393

Table 2. Pairwise Out-of-Sample Forecast Comparison across Expected MAX Models

This table reports pairwise out-of-sample forecast comparisons for alternative models of expected next-month maximum daily returns (MAX). The five forecasting models include a martingale benchmark, an AR(1) model, an ARMA(1,1) model, a quantile–tail approximation (QTAIL), and an EVT-based Peaks-over-Threshold model using the Generalized Pareto Distribution (EVT/GPD). Panel A (t-test) reports the mean difference in RMSE between each pair of models across firms, along with the associated t-statistics in parentheses. Positive values indicate that the row model has higher RMSE (worse performance) than the column model. Panel B (Sign test) reports the percentage of firms for which the column model yields lower RMSE than the row model, with p-values in parentheses testing whether this proportion differs from 50%. Panel C (Wilcoxon signed-rank test) reports nonparametric test statistics for the null hypothesis that the median RMSE difference between the two models is zero, allowing for heavy-tailed forecast-error distributions.

Panel A (t-test)					
	Martingale	AR (1)	ARMA (1,1)	QTAIL	EVT/ GPD
Martingale		88.37 (0.00)	93.69 (0.00)	56.45 (0.00)	55.44 (0.00)
AR (1)	-88.37 (0.00)		29.05 (0.00)	-19.76 (0.00)	-20.00 (0.00)
ARMA (1,1)	-93.69 (0.00)	-29.05 (0.00)		-38.70 (0.00)	-41.59 (0.00)
QTAIL	-56.45 (0.00)	19.76 (0.00)	38.70 (0.00)		3.25 (0.00)
EVT/ GPD	-55.44 (0.00)	20.00 (0.00)	41.59 (0.00)	-3.25 (0.00)	
Panel B (Sign test)					
	Martingale	AR (1)	ARMA (1,1)	QTAIL	EVT/ GPD
Martingale		83.60 (0.00)	85.70 (0.00)	55.16 (0.00)	65.66 (0.00)
AR (1)	-83.60 (0.00)		35.39 (0.00)	-27.76 (0.00)	-36.17 (0.00)
ARMA (1,1)	-85.70 (0.00)	-35.39 (0.00)		-53.69 (0.00)	-64.71 (0.00)
QTAIL	-55.16 (0.00)	27.76 (0.00)	53.69 (0.00)		9.94 (0.00)
EVT/ GPD	-65.66 (0.00)	36.17 (0.00)	64.71 (0.00)	-9.94 (0.00)	
Panel C (Signed-Rank)					
	Martingale	AR (1)	ARMA (1,1)	QTAIL	EVT/ GPD
Martingale		3.34 (0.00)	2.88 (0.00)	8.94 (0.00)	7.35 (0.00)
AR (1)	3.34 (0.00)		14.55 (0.00)	18.50 (0.00)	16.91 (0.00)
ARMA (1,1)	2.88 (0.00)	14.55 (0.00)		10.53 (0.00)	8.31 (0.00)
QTAIL	8.94 (0.00)	18.50 (0.00)	10.53 (0.00)		23.39 (0.00)
EVT/ GPD	7.35 (0.00)	16.91 (0.00)	8.31 (0.00)	23.39 (0.00)	

Table 3. Distributional Forecast Performance and Dependence across Expected MAX Models

This table examines the distributional properties and dependence structure of out-of-sample forecasts for expected next-month maximum daily returns (MAX) across alternative forecasting models. Panel A (Comparison to Best) reports the cross-sectional distribution of percentage performance loss relative to the firm-specific best-performing model. For each firm and month, forecast performance is evaluated using root mean squared error (RMSE), and the best model is defined as the one with the lowest RMSE for that firm. The reported percentiles (p10, p25, p50, p75, p90) summarize how far each model's RMSE lies above the best attainable forecast for a given firm. Lower values indicate closer proximity to the best model and therefore superior and more robust forecasting performance across firms. Panel B (Forecast Correlation) reports average pairwise correlations between forecast series across models. For each firm, correlations are computed using the time series of model-specific expected MAX forecasts, and the reported values represent cross-sectional averages. This panel assesses the degree to which alternative models capture overlapping versus distinct components of extreme-return dynamics. Lower correlations indicate that models exploit different informational sources, while higher correlations suggest common underlying drivers of expected tail behavior.

Panel A: Comparison to best					
Model	p10	p25	p50	p75	p90
Martingale	6.40%	15.71%	25.50%	36.38%	49.65%
AR (1)	0.00%	0.20%	3.41%	8.73%	16.88%
ARMA (1,1)	0.00%	0.00%	0.38%	3.82%	10.58%
QTAIL	0.00%	2.09%	7.75%	17.49%	37.68%
EVT/ GPD	0.00%	3.24%	7.02%	12.69%	22.33%

Panel B: Forecast Correlation					
Model	Martingale	AR (1)	ARMA (1,1)	QTAIL	EVT/ GPD
Martingale	1.00	0.90	0.63	0.40	0.25
AR (1)	0.90	1.00	0.73	0.44	0.33
ARMA (1,1)	0.63	0.73	1.00	0.57	0.48
QTAIL	0.40	0.44	0.57	1.00	0.62
EVT/ GPD	0.25	0.33	0.48	0.62	1.00

Table 4. Expected MAX and Cross-Sectional Return Predictability

This table reports portfolio returns sorted on expected maximum daily returns (EMAX) constructed using alternative forecasting models. At the end of each month, stocks are sorted into quintile portfolios based on expected next-month maximum daily returns. The column labeled Max corresponds to the traditional realized MAX measure (martingale benchmark), while the remaining columns use model-based forecasts: AR (1), ARMA (1,1), Quantile–Tail (QTAIL), and EVT–GPD. Portfolios are held for one month. Panel A reports equal-weighted returns, and Panel B reports value-weighted returns. The row E[re] reports the high-minus-low (H–L) return spread between the highest and lowest expected MAX portfolios, with Newey–West t-statistics in parentheses. The rows FF6 and QF report corresponding alphas from the Fama–French six-factor model and q-factor model, respectively.

Panel A: Equal Weighted					
	Max	EMAX(AR)	EMAX(ARMA)	EMAX(QTAIL)	EMAX(EVT)
Low EMAX	1.25	1.07	1.11	1.23	1.11
2	1.16	1.17	1.21	1.30	1.19
3	1.07	1.31	1.25	1.39	1.21
4	0.92	1.34	1.31	1.41	1.23
High EMAX	0.71	1.32	1.33	1.44	1.27
E[re]	-0.55	0.25	0.23	0.21	0.15
	(-2.40)	(0.80)	(0.36)	(0.35)	(0.33)
FF6	-0.83	-0.25	-0.31	-0.12	-0.23
	(-3.13)	(-0.98)	(-1.05)	(-0.30)	(-0.68)
QF	-0.86	-0.31	-0.40	-0.20	-0.29
	(-3.36)	(-1.32)	(-1.44)	(-0.50)	(-0.86)
Panel B: Value Weighted					
	Max	EMAX(AR)	EMAX(ARMA)	EMAX(QTAIL)	EMAX(EVT)
Low EMAX	1.00	0.90	0.92	1.07	0.93
2	0.99	0.95	1.01	1.16	1.02
3	1.00	1.12	1.00	1.25	0.98
4	0.89	1.09	1.15	1.18	0.94
High EMAX	0.28	0.81	0.69	0.55	0.42
E[re]	-0.72	-0.09	-0.23	-0.52	-0.51
	(-2.67)	(-0.39)	(-1.55)	(-1.50)	(-1.38)
FF6	-0.98	-0.26	-0.18	-0.45	-0.53
	(-3.06)	(-1.10)	(-0.98)	(-1.40)	(-1.53)
QF	-1.04	-0.27	-0.20	-0.50	-0.66
	(-3.24)	(-1.14)	(-1.00)	(-1.50)	(-1.59)

Table 5. Expected MAX and Future Returns: Univariate Fama–MacBeth Regressions

This table reports univariate Fama–MacBeth (1973) cross-sectional regressions of individual stock returns in month $t + 1$ on measures of extreme return behavior observed at the end of month t . In each month, a cross-sectional regression is estimated, and the reported coefficients are the time-series averages of the monthly slope estimates. Newey–West adjusted t -statistics with six lags are reported in parentheses. The column labeled Martingale uses the realized maximum daily return (MAX) in month t as the explanatory variable. The remaining columns replace realized MAX with alternative measures of expected maximum daily returns (E-MAX) constructed using four forecasting approaches: an AR (1) model, an ARMA (1,1) model, a quantile–tail approximation (Tail proxy), and an EVT-based Peaks-over-Threshold model using the Generalized Pareto Distribution (EVT/GPD). Each regression includes an intercept but no additional control variables. The dependent variable is the individual stock’s next-month return.

	Martingale	AR (1)	ARMA (1,1)	Tail proxy	EVT/GPD
Intercept	1.06 (7.33)	0.90 (6.81)	0.92 (7.44)	0.97 (7.85)	0.86 (7.10)
E-MAX	-3.49 (-3.66)	1.77 (0.63)	0.82 (0.32)	0.70 (0.33)	0.86 (0.24)
Adjusted R^2	0.011	0.021	0.024	0.021	0.024

Table 6. Portfolio Returns Sorted on Expected Changes in Maximum Daily Returns (Equal-Weighted)

This table reports equal-weighted portfolio returns formed on the basis of the expected change in maximum daily returns, measured by $D_{i,t} = \text{MAX}_{i,t} - E(\text{MAX}_{i,t+1} | t)$. At the end of each month t , stocks are first classified into two subsamples according to the sign of $D_{i,t}$. Panel A includes stocks with $D_{i,t} > 0$, corresponding to cases in which the model predicts a decline in maximum daily returns in the following month. Panel B includes stocks with $D_{i,t} < 0$, corresponding to cases in which the model predicts an increase in maximum daily returns in the following month. Within each subsample, stocks are sorted into quintile portfolios based on the level of expected maximum daily returns $E(\text{MAX}_{i,t+1})$ constructed using four forecasting models: AR(1), ARMA(1,1), a quantile–tail approximation (QTAIL), and an EVT-based Peaks-over-Threshold model using the Generalized Pareto Distribution (EVT). “Low Dit” (“High Dit”) denotes the lowest (highest) quintile of the sorting variable within each subsample. Reported returns are the average next-month portfolio returns. E[re] reports the difference in average returns between the highest and lowest quintile portfolios. FF6 and QF report risk-adjusted returns estimated using the Fama–French six-factor model and q-factor model, respectively. Corresponding Newey–West adjusted t-statistics are reported in parentheses.

Panel A: Equal weighted - Expected negative surprise ($D_{i,t} > 0$)				
	EMAX(AR)	EMAX(ARMA)	EMAX(QTAIL)	EMAX(EVT)
Low $D_{i,t}$	1.18	1.10	1.28	1.29
2	1.17	1.05	1.29	1.16
3	1.04	1.07	1.21	1.10
4	0.96	0.97	1.00	0.98
High $D_{i,t}$	0.93	0.78	0.81	0.66
E[re]	-0.25	-0.33	-0.47	-0.63
	(-1.27)	(-1.88)	(-2.05)	(-3.50)
FF6	-0.72	-0.77	-0.86	-1.05
	(-3.23)	(-3.69)	(-3.16)	(-4.77)
QF	-0.77	-0.79	-0.92	-1.10
	(-3.68)	(-3.80)	(-3.36)	(-5.20)
Panel B: Equal weighted - Expected positive surprise ($D_{i,t} < 0$)				
	EMAX(AR)	EMAX(ARMA)	EMAX(QTAIL)	EMAX(EVT)
Low $D_{i,t}$	1.18	1.24	1.33	1.29
2	1.28	1.30	1.45	1.37
3	1.37	1.39	1.51	1.44
4	1.44	1.49	1.65	1.40
High $D_{i,t}$	1.47	1.52	1.69	1.64
E[re]	0.28	0.28	0.35	0.35
	(2.11)	(1.70)	(1.43)	(2.10)
FF6	-0.13	-0.16	0.01	0.00
	(-0.99)	(-1.03)	(0.03)	(0.03)
QF	-0.10	-0.15	0.04	0.10
	(-0.75)	(-1.06)	(0.17)	(0.51)

Table 7. Portfolio Returns Sorted on Unexpected Maximum Daily Returns (Equal-Weighted)

This table reports equal-weighted portfolio returns formed by sorting stocks on the unexpected component of maximum daily returns, denoted $UMAX_{i,t+1}$. We define $UMAX_{i,t} = MAX_{i,t+1} - E(MAX_{i,t+1} | t)$. Unexpected MAX is constructed separately using four forecasting models: AR(1), ARMA(1,1), a quantile–tail approximation (QTAIL), and an EVT-based Peaks-over-Threshold model with a Generalized Pareto Distribution (EVT). At the end of each month t , stocks are sorted into quintile portfolios based on $UMAX_{i,t+1}$. Panel A presents results from sorts on the level of unexpected MAX. Panel B restricts the sample to observations with $UMAX_{i,t+1} > 0$, corresponding to months in which realized maximum daily returns exceed their forecasts. Panel C restricts the sample to observations with $UMAX_{i,t+1} < 0$, corresponding to months in which realized maximum daily returns fall short of their forecasts. $E[re]$ reports the difference in average returns between the highest and lowest quintile portfolios. FF6 and QF report risk-adjusted returns estimated using the Fama–French six-factor model and q-factor model, respectively. Newey–West adjusted t-statistics are reported in parentheses.

Panel A: Equal weighted - Unexpected surprise (UMAX)				
	UMAX(AR)	UMAX(ARMA)	UMAX(QTAIL)	UMAX(EVT)
Low UMAX	1.41	1.41	1.24	1.40
2	1.25	1.29	1.06	1.29
3	1.23	1.26	1.03	1.26
4	1.20	1.19	0.98	1.20
High UMAX	1.07	1.00	0.71	0.86
E[re]	-0.34	-0.41	-0.53	-0.54
	(-4.02)	(-5.46)	(-4.69)	(-6.28)
FF6	-0.77	-0.83	-0.92	-0.94
	(-6.26)	(-7.15)	(-6.18)	(-6.33)
QF	-0.80	-0.88	-0.94	-1.04
	(-6.49)	(-7.01)	(-5.80)	(-7.13)
Panel B: Equal weighted - Unexpected negative surprise (UMAX > 0)				
	UMAX(AR)	UMAX(ARMA)	UMAX(QTAIL)	UMAX(EVT)
Low UMAX	1.20	1.15	1.01	1.29
2	1.18	1.08	0.90	1.17
3	1.15	1.07	0.86	1.11
4	1.07	0.99	0.62	0.99
High UMAX	1.02	0.78	0.38	0.66
E[re]	-0.15	-0.36	-0.63	-0.63
	(-0.76)	(-2.00)	(-3.29)	(-3.51)
FF6	-0.61	-0.80	-1.08	-1.06
	(-2.69)	(-3.70)	(-4.39)	(-4.83)
QF	-0.61	-0.82	-1.07	-1.09
	(-2.77)	(-3.78)	(-3.95)	(-5.15)
Panel C: Equal weighted - Unexpected positive surprise (UMAX < 0)				
	UMAX(AR)	UMAX(ARMA)	UMAX(QTAIL)	UMAX(EVT)
Low UMAX	1.20	1.26	0.99	1.28
2	1.28	1.30	1.07	1.37
3	1.33	1.38	1.14	1.43
4	1.43	1.50	1.24	1.46
High UMAX	1.51	1.52	1.29	1.63
E[re]	0.32	0.26	0.30	0.35
	(2.14)	(1.49)	(1.48)	(2.12)
FF6	-0.12	-0.16	-0.09	-0.02
	(-0.86)	(-0.92)	(-0.46)	(-0.08)
QF	-0.11	-0.21	-0.07	0.07
	(-0.88)	(-1.36)	(-0.32)	(0.36)

Table 8. Double-Sorted Portfolio Returns on Unexpected and Realized Maximum Daily Returns (Equal-Weighted)

This table reports equal-weighted portfolio returns from double sorts on the unexpected component of maximum daily returns (UMAX) and realized maximum daily returns (MAX). The unexpected component is defined as $UMAX_{i,t} = MAX_{i,t+1} - E(MAX_{i,t+1} | t)$, where the expected maximum return is obtained from the ARMA(1,1) forecasting model. At the end of each month, stocks are first sorted into quintiles based on UMAX and then into quintiles based on realized MAX, forming 25 portfolios. Panel A reports results for observations with $UMAX_{i,t} > 0$, where realized MAX exceeds its forecast. Panel B reports results for observations with $UMAX_{i,t} < 0$, where realized MAX falls below its forecast. E[re] reports the return difference between the highest and lowest UMAX portfolios within each MAX quintile. α_{FF6} and α_{QF} denote alphas from the Fama–French six-factor model and the q-factor model. Newey–West adjusted t -statistics are reported in parentheses.

Panel A: Equal-Weighted $UMAX > 0 \times MAX$								
	Low MAX	2	3	4	High MAX	E[re]	α_{FF6}	α_{QF}
Low UMAX	1.02	1.08	1.13	1.16	1.24	0.22 (0.81)	-0.20 (-0.77)	-0.18 (-0.69)
2	1.00	1.10	1.05	1.12	1.12	0.13 (0.51)	-0.26 (-1.04)	-0.40 (-1.64)
3	0.98	1.08	1.01	1.09	1.09	0.11 (0.41)	-0.27 (-1.15)	-0.32 (-0.99)
4	0.94	0.98	1.11	1.08	0.84	-0.10 (-0.30)	-0.52 (-1.47)	-0.64 (-1.91)
High UMAX	0.85	0.81	0.79	0.66	0.54	-0.31 (-0.83)	-0.90 (-2.38)	-0.92 (-2.05)
E[re]	-0.17 (-1.01)	-0.28 (-1.47)	-0.34 (-1.89)	-0.50 (-2.09)	-0.70 (-2.34)			
α_{FF6}	-0.58 (-3.27)	-0.62 (-2.69)	-0.58 (-1.51)	-1.01 (-3.44)	-1.28 (-4.02)			
α_{QF}	-0.56 (-2.77)	-0.68 (-3.01)	-0.53 (-1.34)	-1.12 (-3.61)	-1.30 (-3.34)			
Panel B: Equal-Weighted $UMAX < 0 \times MAX$								
	Low MAX	2	3	4	High MAX	E[re]	α_{FF6}	α_{QF}
Low UMAX	1.11	1.24	1.23	1.24	1.43	0.31 (1.33)	-0.05 (-0.20)	0.03 (0.11)
2	1.16	1.27	1.33	1.42	1.41	0.25 (1.06)	-0.03 (-0.10)	-0.09 (-0.38)
3	1.19	1.36	1.49	1.52	1.36	0.17 (0.69)	-0.19 (-0.80)	-0.14 (-0.55)
4	1.19	1.55	1.53	1.55	1.61	0.42 (1.50)	0.16 (0.51)	-0.18 (-0.74)
High UMAX	1.32	1.57	1.57	1.65	1.54	0.22 (0.63)	-0.18 (-0.52)	-0.36 (-1.04)
E[re]	0.20 (1.40)	0.33 (1.90)	0.35 (1.66)	0.41 (1.93)	0.11 (0.40)			
α_{FF6}	-0.21 (-1.38)	-0.07 (-0.42)	-0.04 (-0.20)	-0.00 (-0.17)	-0.35 (-1.16)			
α_{QF}	-0.14 (-0.83)	-0.10 (-0.54)	-0.12 (-0.59)	-0.03 (-0.40)	-0.53 (-1.76)			

Table 9. Portfolio Returns from Double Sorts on Unexpected and Expected Maximum Daily Returns (Equal-Weighted)

This table reports equal-weighted portfolio returns from double sorts on the unexpected component of maximum daily returns (UMAX) and the expected maximum daily return (EMAX), where EMAX is obtained from the ARMA(1,1) forecasting model and $UMAX_{i,t} = MAX_{i,t+1} - E(MAX_{i,t+1} | t)$. At the end of each month, stocks are first sorted into quintiles based on UMAX and then into quintiles based on EMAX, forming 25 portfolios. Panel A reports results for observations with $UMAX_{i,t} > 0$ (realized MAX exceeds its forecast), while Panel B reports results for observations with $UMAX_{i,t} < 0$ (realized MAX falls below its forecast). E[re] reports the return difference between the highest and lowest UMAX portfolios within each EMAX quintile. α_{FF6} and α_{QF} denote alphas from the Fama–French six-factor model and the q-factor model. Newey–West adjusted t -statistics are reported in parentheses.

Panel A: Equal-Weighted $UMAX > 0 \times EMAX$								
	Low EMAX	2	3	4	High EMAX	E[re]	α_{FF6}	α_{QF}
Low UMAX	1.04	1.06	1.19	1.20	1.22	0.19 (0.62)	-0.25 (-0.93)	-0.23 (-0.88)
2	1.00	1.06	1.12	1.12	1.14	0.14 (0.41)	-0.30 (-1.20)	-0.42 (-1.75)
3	1.07	1.06	1.09	1.04	1.08	0.01 (0.04)	-0.37 (-1.15)	-0.30 (-0.95)
4	1.00	0.92	1.15	1.11	0.78	-0.22 (-0.73)	-0.66 (-2.09)	-0.83 (-2.70)
High UMAX	0.82	0.80	0.75	0.71	0.47	-0.35 (-0.94)	-0.89 (-2.36)	-0.95 (-2.28)
E[re]	-0.21 (-1.51)	-0.26 (-1.71)	-0.44 (-1.77)	-0.49 (-1.98)	-0.75 (-2.31)			
α_{FF6}	-0.57 (-3.68)	-0.66 (-2.50)	-0.92 (-3.02)	-1.02 (-2.05)	-1.21 (-3.78)			
α_{QF}	-0.57 (-3.57)	-0.65 (-2.21)	-0.87 (-3.05)	-0.94 (-2.42)	-1.29 (-3.36)			
Panel B: Equal-Weighted $UMAX < 0 \times EMAX$								
	Low EMAX	2	3	4	High EMAX	E[re]	α_{FF6}	α_{QF}
Low UMAX	1.13	1.21	1.23	1.28	1.41	0.28 (1.20)	-0.10 (-0.42)	-0.02 (-0.09)
2	1.14	1.18	1.34	1.39	1.44	0.30 (1.24)	0.01 (0.02)	-0.05 (-0.20)
3	1.22	1.36	1.46	1.49	1.53	0.31 (1.36)	-0.19 (-0.07)	-0.14 (-0.17)
4	1.19	1.53	1.54	1.76	1.60	0.42 (1.52)	0.21 (0.68)	-0.13 (-0.51)
High UMAX	1.41	1.55	1.53	1.61	1.42	0.01 (0.04)	-0.26 (-0.88)	-0.49 (-1.58)
E[re]	0.28 (1.61)	0.34 (1.75)	0.30 (1.55)	0.32 (1.37)	0.02 (0.07)			
α_{FF6}	-0.17 (-0.96)	-0.15 (-0.81)	-0.09 (-0.42)	0.01 (0.06)	-0.33 (-1.30)			
α_{QF}	-0.14 (-0.78)	-0.10 (-0.51)	-0.10 (-0.56)	-0.02 (-0.09)	-0.60 (-2.55)			

Table 10. Bivariate and Multivariate Fama–MacBeth Regressions with Realized, Expected, and Unexpected MAX

This table reports monthly Fama–MacBeth (1973) cross-sectional regression results examining the relation between next-month stock returns and realized, expected, and unexpected components of maximum daily returns. The dependent variable in all specifications is the stock's return in month $t + 1$, denoted $R_{i,t+1}$. The regressions include alternative measures of expected maximum daily returns ($EMAX_{i,t}$) constructed using five forecasting approaches: a martingale benchmark, AR (1), ARMA (1,1), a quantile–tail proxy, and an EVT-based Peaks-over-Threshold model with a Generalized Pareto Distribution (EVT/GPD). The unexpected component of maximum daily returns ($UMAX_{i,t+1}$) is defined as the realized maximum daily return in month $t + 1$ minus its forecast formed at the end of month t . Panel A reports bivariate regressions of next-month returns on realized MAX and expected MAX. Panel B reports bivariate regressions of next-month returns on expected MAX and unexpected MAX. Panel C reports bivariate regressions of next-month returns on realized MAX and unexpected MAX. Each column corresponds to a different expected-MAX forecasting model. Reported coefficients are time-series averages of monthly cross-sectional estimates. Newey–West adjusted t-statistics are reported in parentheses.

Panel A: $R_{i,t+1} = \alpha + \beta MAX_{i,t} + \gamma EMAX_{i,t} + \varepsilon_{i,t}$					
	Martingale	AR (1)	ARMA (1,1)	Tail proxy	EVT/GPD
Intercept	1.06 (7.33)	0.81 (6.27)	0.87 (7.26)	1.00 (8.06)	0.87 (7.14)
MAX	-3.49 (-3.66)	-5.12 (-5.58)	-4.74 (-6.55)	-4.83 (-7.14)	-4.40 (-7.87)
EMAX		8.88 (2.64)	6.49 (2.35)	4.59 (2.30)	6.40 (1.88)
Adjusted R^2	0.011	0.024	0.026	0.023	0.026
Panel B: $R_{i,t+1} = \alpha + \beta EMAX_{i,t} + \gamma UMAX_{i,t} + \varepsilon_{i,t}$					
	Martingale	AR (1)	ARMA (1,1)	Tail proxy	EVT/GPD
Intercept	1.05 (8.11)	0.80 (6.39)	0.84 (7.03)	0.95 (6.01)	0.86 (7.15)
EMAX	-3.57 (-2.64)	3.86 (1.28)	2.29 (0.86)	0.28 (0.11)	2.22 (0.63)
UMAX	-3.46 (-4.66)	-3.75 (-4.79)	-4.47 (-7.33)	-3.94 (-5.58)	-4.33 (-8.01)
Adjusted R^2	0.016	0.025	0.026	0.024	0.026
Panel C: $R_{i,t+1} = \alpha + \beta MAX_{i,t} + \gamma UMAX_{i,t} + \varepsilon_{i,t}$					
	Martingale	AR (1)	ARMA (1,1)	Tail proxy	EVT/GPD
Intercept	1.05 (8.11)	0.86 (7.05)	0.87 (7.37)	0.77 (6.03)	0.85 (7.10)
MAX	-3.57 (-2.64)	2.41 (0.84)	1.52 (0.57)	-1.16 (-0.56)	2.52 (0.71)
UMAX	0.11 (0.16)	-5.06 (-2.28)	-5.26 (-2.51)	-3.24 (-1.83)	-6.85 (-2.04)
Adjusted R^2	0.016	0.025	0.026	0.024	0.026

Table 11. Multivariate Fama–MacBeth Cross-Sectional Regressions

This table reports results from monthly Fama–MacBeth (1973) cross-sectional regressions examining the relation between next-month stock returns and expected and unexpected components of maximum daily returns, controlling for standard firm characteristics. The dependent variable in all specifications is the stock’s return in month $t + 1$, denoted $R_{i,t+1}$. The key explanatory variables are the expected maximum daily return ($EMAX_{i,t}$) and the unexpected maximum daily return ($UMAX_{i,t+1}$). Expected MAX is constructed using five alternative forecasting approaches: a martingale benchmark, AR (1), ARMA (1,1), a quantile–tail proxy, and an EVT-based Peaks-over-Threshold model with a Generalized Pareto Distribution (EVT/GPD). Unexpected MAX is defined as the realized maximum daily return in month $t + 1$ minus its forecast formed at the end of month t . All regressions additionally control for market beta (BETA), firm size (log market capitalization), book-to-market ratio (BTM), momentum (MOM), short-term reversal (REVERSAL), illiquidity (ILLIQUIDITY), idiosyncratic volatility (IVOL), and idiosyncratic skewness (ISKEW). All control variables are measured at the end of month t . Reported coefficients are time-series averages of the monthly cross-sectional estimates. Newey–West adjusted t-statistics are reported in parentheses.

	Martingale	AR (1)	ARMA (1,1)	Tail proxy	EVT/GPD
Intercept	2.54 (7.49)	2.04 (7.22)	2.47 (9.65)	2.38 (7.53)	2.43 (8.77)
E-MAX	-4.58 (-3.49)	-3.88 (-1.27)	-1.86 (-0.63)	-2.03 (-0.81)	-2.79 (-0.68)
U-MAX	-5.96 (-7.50)	-4.12 (-3.35)	-5.85 (-6.47)	-3.99 (-5.77)	-5.59 (-7.03)
BETA	0.00 (0.15)	0.00 (0.11)	0.01 (0.48)	0.01 (0.54)	0.00 (0.10)
Size	-0.14 (-4.73)	-0.12 (-4.84)	-0.13 (-5.35)	-0.15 (-4.77)	-0.13 (-4.89)
BTM	0.05 (2.42)	0.04 (2.71)	0.04 (1.72)	0.04 (1.36)	0.04 (2.09)
MOM	0.48 (3.40)	0.55 (3.58)	0.47 (3.47)	0.59 (5.11)	0.46 (3.35)
REVERSAL	-0.56 (-10.95)	-0.56 (-12.57)	-0.60 (-12.36)	-0.55 (-9.34)	-0.58 (-11.93)
ILLIQUIDTY	0.04 (2.00)	0.04 (2.26)	0.04 (2.80)	0.03 (1.64)	0.03 (2.33)
IVOL	0.62 (2.55)	0.21 (1.57)	0.54 (2.50)	0.49 (1.61)	0.58 (2.07)
ISKEW	-0.02 (-1.06)	-0.03 (-1.69)	0.00 (-0.23)	-0.04 (-2.28)	-0.03 (-2.10)
Adjusted R^2	0.047	0.059	0.055	0.050	0.052

Appendix A

Table 6B. Portfolio Returns Sorted on Expected Changes in Maximum Daily Returns (Value-Weighted)

This table reports value-weighted portfolio returns formed on the basis of the expected change in maximum daily returns, defined as $D_{i,t} = MAX_{i,t} - E(MAX_{i,t+1} | t)$. Stocks are first classified according to the sign of $D_{i,t}$. Panel A reports results for observations with $D_{i,t} > 0$, where the model predicts a decline in maximum daily returns in the following month. Panel B reports results for observations with $D_{i,t} < 0$, where the model predicts an increase in maximum daily returns. Within each subsample, stocks are sorted into quintiles based on the expected maximum daily return $E(MAX_{i,t+1})$ estimated using AR(1), ARMA(1,1), QTAIL, and EVT models. E[re] reports the return difference between the highest and lowest quintile portfolios. FF6 and QF denote alphas from the Fama–French six-factor model and the q-factor model. Newey–West adjusted t -statistics are reported in parentheses.

Panel A: Value weighted - Expected negative surprise ($D_{i,t} > 0$)				
	EMAX(AR)	EMAX(ARMA)	EMAX(QTAIL)	EMAX(EVT)
Low $D_{i,t}$	0.93	0.81	1.13	0.93
2	0.81	0.80	1.11	0.91
3	0.95	0.75	1.08	0.86
4	0.87	0.71	0.99	0.84
High $D_{i,t}$	0.74	0.66	0.83	0.42
E[re]	-0.19	-0.15	-0.31	-0.51
	(-1.00)	(-0.84)	(-1.48)	(-2.72)
FF6	-0.66	-0.54	-0.71	-0.96
	(-2.69)	(-2.07)	(-2.46)	(-3.55)
QF	-0.84	-0.69	-0.85	-1.06
	(-3.06)	(-2.24)	(-2.62)	(-3.68)
Panel B: Value weighted - Expected positive surprise ($D_{i,t} < 0$)				
	EMAX(AR)	EMAX(ARMA)	EMAX(QTAIL)	EMAX(EVT)
Low $D_{i,t}$	0.98	0.96	1.04	0.94
2	0.93	1.02	1.08	1.05
3	1.03	1.06	1.18	1.14
4	1.28	1.18	1.24	1.15
High $D_{i,t}$	1.25	1.23	1.28	1.14
E[re]	0.27	0.28	0.25	0.20
	(1.70)	(1.75)	(1.33)	(1.13)
FF6	-0.10	-0.01	-0.02	-0.12
	(-0.61)	(-0.06)	(-0.12)	(-0.58)
QF	-0.05	-0.06	-0.10	-0.07
	(-0.30)	(-0.31)	(-0.41)	(-0.30)

Table 7B. Portfolio Returns Sorted on Unexpected Maximum Daily Returns (Value-Weighted)

This table reports value-weighted portfolio returns formed by sorting stocks on the unexpected component of maximum daily returns (UMAX). The unexpected component is defined as $UMAX_{i,t} = MAX_{i,t+1} - E(MAX_{i,t+1} | t)$. Expected maximum returns are obtained from four forecasting models: AR(1), ARMA(1,1), a quantile–tail approximation (QTAIL), and an EVT-based Peaks-over-Threshold model with a Generalized Pareto Distribution (EVT). At the end of each month, stocks are sorted into quintile portfolios based on UMAX. Panel A reports results for the full sample. Panel B restricts the sample to observations with $UMAX_{i,t} > 0$, where realized MAX exceeds its forecast. Panel C restricts the sample to observations with $UMAX_{i,t} < 0$, where realized MAX falls below its forecast. E[re] reports the return difference between the highest and lowest UMAX portfolios. α_{FF6} and α_{QF} denote alphas from the Fama–French six-factor model and the q-factor model. Newey–West adjusted t -statistics are reported in parentheses.

Panel A: Value weighted - Unexpected surprise (UMAX)				
	UMAX(AR)	UMAX(ARMA)	UMAX(QTAIL)	UMAX(EVT)
Low UMAX	1.06	1.07	0.86	1.09
2	0.94	0.99	0.82	0.95
3	0.92	0.96	0.75	0.98
4	0.99	0.94	0.68	0.89
High UMAX	0.84	0.86	0.60	0.64
E[re]	-0.22	-0.21	-0.26	-0.44
	(-1.65)	(-1.83)	(-2.09)	(-2.98)
α_{FF6}	-0.71	-0.65	-0.52	-0.89
	(-4.12)	(-3.94)	(-2.24)	(-4.14)
α_{QF}	-0.76	-0.78	-0.58	-1.05
	(-4.61)	(-4.13)	(-2.01)	(-4.38)
Panel B: Value weighted - Unexpected negative surprise (UMAX>0)				
	UMAX(AR)	UMAX(ARMA)	UMAX(QTAIL)	UMAX(EVT)
Low UMAX	0.96	0.94	0.92	0.94
2	0.90	0.88	0.75	0.92
3	0.87	0.83	0.70	0.86
4	0.87	0.74	0.65	0.86
High UMAX	0.84	0.69	0.52	0.43
E[re]	-0.12	-0.25	-0.40	-0.52
	(-0.89)	(-1.30)	(-1.89)	(-2.69)
α_{FF6}	-0.48	-0.56	-0.82	-0.96
	(-1.88)	(-2.12)	(-3.29)	(-3.52)
α_{QF}	-0.69	-0.71	-0.99	-1.05
	(-2.26)	(-2.33)	(-3.71)	(-3.60)
Panel C: Value weighted - Unexpected positive surprise (UMAX<0)				
	UMAX(AR)	UMAX(ARMA)	UMAX(QTAIL)	UMAX(EVT)
Low UMAX	0.88	0.96	0.85	0.97
2	0.92	1.01	0.88	1.05
3	1.12	1.02	0.94	1.13
4	1.15	1.17	0.96	1.16
High UMAX	1.21	1.22	1.03	1.19
E[re]	0.33	0.26	0.18	0.22
	(2.51)	(1.55)	(1.06)	(1.13)
α_{FF6}	-0.01	-0.08	-0.12	-0.11
	(-0.07)	(-0.46)	(-1.08)	(-0.79)
α_{QF}	-0.02	-0.13	-0.46	-0.10
	(-0.13)	(-0.69)	(-1.07)	(-0.41)

Table 8B. Double-Sorted Portfolio Returns on Unexpected and Realized Maximum Daily Returns (Value-Weighted)

This table reports value-weighted portfolio returns from double sorts on the unexpected component of maximum daily returns (UMAX) and realized maximum daily returns (MAX). The unexpected component is defined as $UMAX_{i,t} = MAX_{i,t+1} - E(MAX_{i,t+1} | t)$, where the expected maximum return is obtained from the ARMA(1,1) forecasting model. At the end of each month, stocks are first sorted into quintiles based on UMAX and then into quintiles based on MAX, forming 25 portfolios. Panel A reports results for observations with $UMAX_{i,t} > 0$, where realized MAX exceeds its forecast. Panel B reports results for observations with $UMAX_{i,t} < 0$, where realized MAX falls below its forecast. E[re] reports the return difference between the highest and lowest UMAX portfolios within each MAX quintile. α_{FF6} and α_{QF} denote alphas from the Fama–French six-factor model and the q-factor model. Newey–West adjusted t -statistics are reported in parentheses.

Panel A: Value-Weighted $UMAX > 0 \times MAX$								
	Low MAX	2	3	4	High MAX	E[re]	α_{FF6}	α_{QF}
Low UMAX	0.84	0.82	1.02	1.06	0.83	-0.01 (-0.02)	-0.22 (-0.71)	-0.07 (-0.22)
2	0.77	0.84	0.83	0.93	0.64	-0.14 (-0.43)	-0.20 (-0.60)	-0.38 (-1.02)
3	1.05	1.08	0.62	0.82	0.59	-0.45 (-1.38)	-0.86 (-2.34)	-0.87 (-2.05)
4	1.00	0.69	0.89	0.76	0.03	-0.97 (-2.72)	-1.33 (-3.25)	-1.56 (-3.37)
High UMAX	0.74	0.66	0.55	0.24	-0.77	-1.51 (-4.08)	-1.96 (-5.12)	-1.88 (-4.63)
E[re]	-0.10 (-0.58)	-0.16 (-0.58)	-0.47 (-1.46)	-0.82 (-2.36)	-1.60 (-5.09)			
α_{FF6}	-0.49 (-1.86)	-0.51 (-1.54)	-0.87 (-2.27)	-1.23 (-3.13)	-2.23 (-6.28)			
α_{QF}	-0.59 (-1.87)	-0.72 (-2.02)	-0.93 (-2.20)	-1.61 (-3.78)	-2.40 (-6.58)			
Panel B: Value-Weighted $UMAX < 0 \times MAX$								
	Low MAX	2	3	4	High MAX	E[re]	α_{FF6}	α_{QF}
Low UMAX	0.93	1.02	1.05	1.10	1.21	0.28 (1.09)	0.13 (0.47)	0.22 (0.87)
2	1.03	0.90	1.03	1.31	0.87	-0.16 (-0.53)	-0.16 (-0.51)	-0.13 (-0.40)
3	1.09	1.00	1.05	1.09	1.16	0.06 (0.21)	-0.14 (-0.51)	-0.12 (-0.42)
4	1.13	1.32	1.53	1.35	0.83	-0.31 (-1.03)	-0.55 (-1.81)	-0.59 (-1.73)
High UMAX	1.35	1.23	1.04	0.60	0.55	-0.81 (-2.82)	-1.26 (-4.08)	-1.56 (-4.91)
E[re]	0.42 (2.57)	0.21 (1.17)	-0.01 (-0.03)	-0.51 (-1.76)	-0.66 (-3.07)			
α_{FF6}	0.18 (0.95)	-0.17 (-0.83)	-0.24 (-1.03)	-0.70 (-2.34)	-1.20 (-4.55)			
α_{QF}	0.17 (0.89)	-0.27 (-1.06)	-0.26 (-1.17)	-0.75 (-2.19)	-1.61 (-5.63)			

Table 9B. Double-Sorted Portfolio Returns on Unexpected and Expected Maximum Daily Returns (Value-Weighted)

This table reports value-weighted portfolio returns from double sorts on the unexpected component of maximum daily returns (UMAX) and the expected maximum daily return (EMAX), where EMAX is obtained from the ARMA(1,1) forecasting model and $UMAX_{i,t} = MAX_{i,t+1} - E(MAX_{i,t+1} | t)$. At the end of each month, stocks are first sorted into quintiles based on UMAX and then into quintiles based on EMAX, forming 25 portfolios. Panel A reports results for observations with $UMAX_{i,t} > 0$ (realized MAX exceeds its forecast), while Panel B reports results for observations with $UMAX_{i,t} < 0$ (realized MAX falls below its forecast). E[re] reports the return difference between the highest and lowest UMAX portfolios within each EMAX quintile. α_{FF6} and α_{QF} denote alphas from the Fama–French six-factor model and the q-factor model. Newey–West adjusted t -statistics are reported in parentheses.

Panel A: Value -Weighted $UMAX > 0 \times EMAX$								
	Low EMAX	2	3	4	High EMAX	E[re]	α_{FF6}	α_{QF}
Low UMAX	0.90	0.94	0.96	1.10	1.13	0.23 (0.76)	-0.06 (-0.12)	-0.01 (-0.18)
2	0.75	0.89	0.83	0.91	0.63	-0.12 (-0.37)	-0.29 (-0.86)	-0.46 (-1.16)
3	0.70	0.77	0.83	0.62	0.65	-0.43 (-1.29)	-0.81 (-2.07)	-0.84 (-1.94)
4	0.67	0.64	0.72	0.54	0.09	-0.57 (-1.81)	-0.86 (-1.80)	-0.96 (-2.15)
High UMAX	0.64	0.61	0.20	0.14	0.01	-0.63 (-2.22)	-1.22 (-2.34)	-1.52 (-3.07)
E[re]	-0.26 (-0.91)	-0.33 (-1.32)	-0.76 (-2.76)	-0.96 (-5.03)	-1.12 (-5.53)			
α_{FF6}	-0.36 (-1.76)	-0.57 (-1.23)	-1.13 (-3.22)	-1.50 (-4.28)	-1.81 (-4.61)			
α_{QF}	-0.50 (-1.61)	-0.56 (-1.30)	-1.30 (-3.54)	-2.03 (-4.88)	-2.36 (-5.03)			
Panel B: Value -Weighted $UMAX < 0 \times EMAX$								
	Low EMAX	2	3	4	High EMAX	E[re]	α_{FF6}	α_{QF}
Low UMAX	0.91	1.06	1.05	1.16	1.07	0.16 (0.65)	-0.01 (-0.03)	0.08 (0.31)
2	0.99	0.90	1.10	1.27	0.88	-0.11 (-0.37)	-0.09 (-0.29)	-0.09 (-0.26)
3	1.15	1.02	1.08	1.02	1.02	-0.13 (-0.45)	-0.30 (-1.07)	-0.33 (-1.19)
4	1.16	1.19	1.51	1.43	0.85	-0.31 (-1.19)	-0.48 (-1.75)	-0.61 (-2.06)
High UMAX	1.06	1.06	1.30	0.74	0.59	-0.46 (-1.41)	-0.67 (-1.86)	-0.73 (-1.76)
E[re]	0.15 (0.77)	0.00 (0.01)	0.25 (1.26)	-0.42 (-1.81)	-0.48 (-2.16)			
α_{FF6}	-0.22 (-1.00)	-0.39 (-1.82)	-0.15 (-0.66)	-0.52 (-2.01)	-0.88 (-3.05)			
α_{QF}	-0.30 (-1.36)	-0.34 (-1.31)	-0.11 (-0.44)	-0.79 (-2.65)	-1.11 (-3.62)			

